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Wing Cheung

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Exploring parsimonious principles that unify active portfolio selection (I): model

WING CHEUNG †‡§*

†Lehman Brothers, 25 Bank Street, London E14 5LE, UK;
‡Nomura International plc, 1 Angel Lane, London EC4R 3AB, UK

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This research unifies *active* portfolio selection through (1) *Subjective Allocation Rule* (SAR)-based allocation and (2) *Minimum Tracking Error* (MTE)-guided mimicking within a Bayesian Allocation Framework that integrates qualitative and quantitative beliefs into alpha-generating portfolios. Portfolio selection is modelled as *view-driven, feedback-responsive* rationality that reconciles *normative* and *descriptive* paradigms, with diverse investor behaviours emerging as *local optima* shaped by cognition, skill, and subjectivity. Empirical results show that when *subjectivity* aligns with predictive *skill*, portfolios outperform mean-variance benchmarks. This first paper focuses on the *structural formulation*.

Keywords: Active portfolio selection; Bounded rationality; Factor mimicking and hedging; Forecasting skill and error; Ambiguity and subjectivity; Markowitz

JEL Classifications: C38, C61, D81, G11, G14, G41

1. Introduction

Traditional mean-variance (MV) portfolio optimisation relies on precise estimates of returns and risks, which can be high-dimensional and inherently error-prone. In practice, cognitively bounded investors tend to prefer simpler, more intuitive approaches: (1) *dimension reduction*, leveraging views on a sparse and selective few factors or themes to guide broad portfolios; (2) *integration of diverse information*, including quantitative, qualitative, objective, or subjective inputs that humans can naturally provide; and (3) *scalable, easy-to-apply mechanisms* for managing complex objectives. These preferences, noted by Salo *et al.* (2024), underlie many common industry portfolio-selection (PS) behaviours¶ – such as *equal weighting*, *factor ranking*, *portfolio algebra*, *volatility-managed portfolios*, *hedging*, and *portable alpha*‖ – which remain inadequately explained or unified within Markowitz (1952) portfolio theory (MPT), its extensions, or any alternative framework.

Unifying theory with real-world behaviour is not just academic – it is essential in practice. Without it, fund managers rely on *ad hoc* methods, leading to portfolios constructed under inconsistent principles; regulators and stakeholders lack coherent standards for assessing fund management outcomes; and robo-advisors struggle to coherently personalise portfolios in real time for different investor circumstances, preferences, and insights.

This research aims to identify the core principles of *active portfolio selection* that unify diverse investor behaviours, while examining their underlying economic mechanisms. It systematically decodes the often-overlooked role of investor *subjectivity* in enhancing performance – complementing

*Corresponding author. Email: W.Cheung.02@cantab.net
§Current affiliation: Department of Finance, University of Roehampton, London, UK

¶In this research, to reflect the *descriptive* modelling paradigm, *behaviour* refers to portfolio-decision patterns directly observable among fund managers, distinct from psychological constructs (e.g. Prospect Theory, Mental Accounting, SP/A theory) typically used in *normative* models (see Footnote 12 for examples).

‖*Equal weighting* (often referred to as '1/N'), as seen in equal-weighted indices and various factor strategies, allocates an equal percentage weight to each asset, irrespective of its market capitalisation, risk, or other characteristics. *Factor ranking* is a factor investing technique that uses Fama-French ranking (Fama and French,1992, 1993) to gain factor exposures. *Portfolio algebra* refers to the combination of strategies through simple linear algebraic operations (i.e. summation, deduction, and multiplication) applied to portfolio weight vectors. It is commonly used to combine ranking-based factor styles. A *volatility-managed portfolio*, as described in Moreira and Muir (2017), allocates capital based on the return-to-variance ratio. A *portable alpha* strategy separates *alpha* (i.e. active returns, or outperformance, from active management) from *beta* (passive market returns), allowing alpha from one asset class or strategy to be overlaid onto beta from another. This research explores the fundamental principles that underpin and unify these behaviours.

investor *skill*† as the key driver – and challenges the conventional view that success depends mainly on estimation accuracy.

Treynor and Black (1973) note that MPT may not be necessary for *passive* investors, who should simply hold a market index portfolio according to the Capital Asset Pricing Model (CAPM). However, MPT may also be insufficient for *active* investors, as it does not flexibly accommodate investor-specific insights. They therefore highlight the need to reconcile *subjective* analyst views with the *objective* discipline of portfolio construction – to establish guiding principles that coherently integrate diverse and heterogeneous investor views within a unified PS framework.

Today, despite the abundance of view- or alpha-generation models‡ and PS techniques§, the principles unifying their connections remain unclear. Black and Litterman (1992) uses Bayes' Rule to tilt the market view towards the investor's subjective opinions, bridging *passive* and *active* investment styles. However, it does not fully integrate the diverse views investors hold, largely due to cognitive limitations and differing interpretation of the market. For instance, in equity portfolio management, investor views are often expressed not only on *individual stocks*¶ or *stock baskets (or themes)*‖ (which the BL model can incorporate), but also on *factors*∗∗ or *stock-specific returns*†† (which it does not). In practice, these views are implemented *separately* through strategies like *stock-picking*, *thematic investing*, *quant investing*, or *portable alpha*, without a unified framework.

The Augmented Black-Litterman (ABL) Model (Cheung 2013) extends BL by unifying all the above views within a coherent framework, elegantly linking subjective opinions with objective portfolio construction in a logically consistent and operationally simple way. Studies‡‡ suggest that ABL can produce robust portfolios with targeted active exposures. Leveraging its unification feature, this paper investigates its inner workings to illuminate fundamental active PS principles.

A **mechanism dissection methodology** is introduced to identify the key functional components and their symmetries from the closed-form ABL allocation formula, which unifies *strategy combination*, *factor mimicking*, *hedging*, and *portable alpha*. From these components, the methodology extracts the underlying efficiency principles and economic mechanisms. Each principle and mechanism is then empirically validated, extending the insights beyond ABL to broader portfolio theory.

Figure 1 illustrates the full research roadmap, with the dashed line separating theoretical and empirical components. For conciseness, the research is presented as a two-part series. This *theoretical* paper, Paper (I), derives the principles of active PS and their underlying economic mechanisms (above the dashed line), illustrated with a worked example. Section 2 introduces the mechanism-dissection methodology. Section 3 derives the closed-form ABL allocation. Subsection 4.1 and 4.2 extract the efficient *mimicking* and *allocation* principles, unifying internal mimicking and hedging, and view blending and strategy combination, respectively. Subsections 4.3 and 4.4 examine the alpha-generation and cognition-friendly mechanisms of the *allocation* principle using a novel **descriptive characterisation methodology**. Subsection 4.5 discusses parsimony and completeness, Section 5 presents a worked example, and Section 6 explores implications for *active* PS practices.

The next *empirical* paper, Paper (II) (Cheung 2024), validates these principles (below the dashed line) and introduces specific methodologies to evaluate their underlying economic mechanisms, demonstrating how investor *skill* and *subjectivity* affect performance.

Therefore, rather than simply extending previous models, this research uncovers the core principles of active PS theory and practice by: (1) dissecting a model that unifies theory and practice to reveal its inner workings; (2) extracting and validating its guiding principles, along with their economic mechanisms; and (3) identifying the key factors that drive portfolio performance.

2. The mechanism dissection methodology

While a complex model exhibits desirable properties not explained by existing theories, its effectiveness is likely driven by hidden guiding principles. *Mechanism dissection* uncovers these principles by analysing its components and interactions, much like an autopsy reveals an organism's inner workings. This methodology yields insights that enhance theory, explain observed phenomena, and inform practical applications.

The ABL model, which uniquely unifies diverse investor behaviours not captured in existing portfolio theories,

† See Subsection 4.3 for the technical definition of *skill*.

‡ In addition to various *moving average* and *econometric* models, there is a rich *factor-based return anomaly modelling* literature, e.g. Fama and French (1992, 1993, 1996), Carhart (1997), Cochrane (1999), Feng *et al.* (2020), and Hou *et al.* (2020); and those reviewed in Goyal *et al.* (2023).

§ In addition to Markowitz (1952), there is a rich *portfolio construction* or *portfolio selection* literature, e.g. Sharpe (1963), Elton *et al.* (1976), Best and Grauer (1991), Goldfarb and Iyengar (2003), Garlappi *et al.* (2007), Kolm *et al.* (2014), Ao *et al.* (2019), and Blanchet *et al.* (2022).

¶ Equity analysts follow a number of individual stocks and form their *stock-picking* strategies.

‖ For example, *pair-trading views* explore the relative performance of one stock versus another; and *thematic views* invest into a basket of stocks representing an investment theme, including *quant style* baskets (e.g. Size, Value, Momentum) based on Fama-French factor rankings, inspired by Fama and French (1992, 1993, 1996).

∗∗ *Factor views* have traditionally been implemented as quant styles through factor rankings. Declining style performance, crowded trades (e.g. Cheung and Mishra (2010a) and DeMiguel *et al.* (2021), and the need for holistic factor management complicate this approach. Alternatively, views can be expressed through factors generated by a multifactor model. Accordingly, this paper refers to predefined ranking-based baskets as factor '*baskets*' or '*themes*', while model-generated factors are simply called '*factors*'.

†† *Stock-specific betting* generates *portable alpha*, independent of broader market movements, which can be applied or 'ported' to other benchmarks. To achieve this, investors typically use index futures to hedge systematic risk, raising questions about efficiency.

‡‡ See for example, Meucci (2009), Luo and Hu (2011), Dion *et al.* (2012), Yang *et al.* (2012), Walters (2014), Dewandaru *et al.* (2015), Kim *et al.* (2017), and Palczewski and Palczewski (2017, 2019) for discussions.

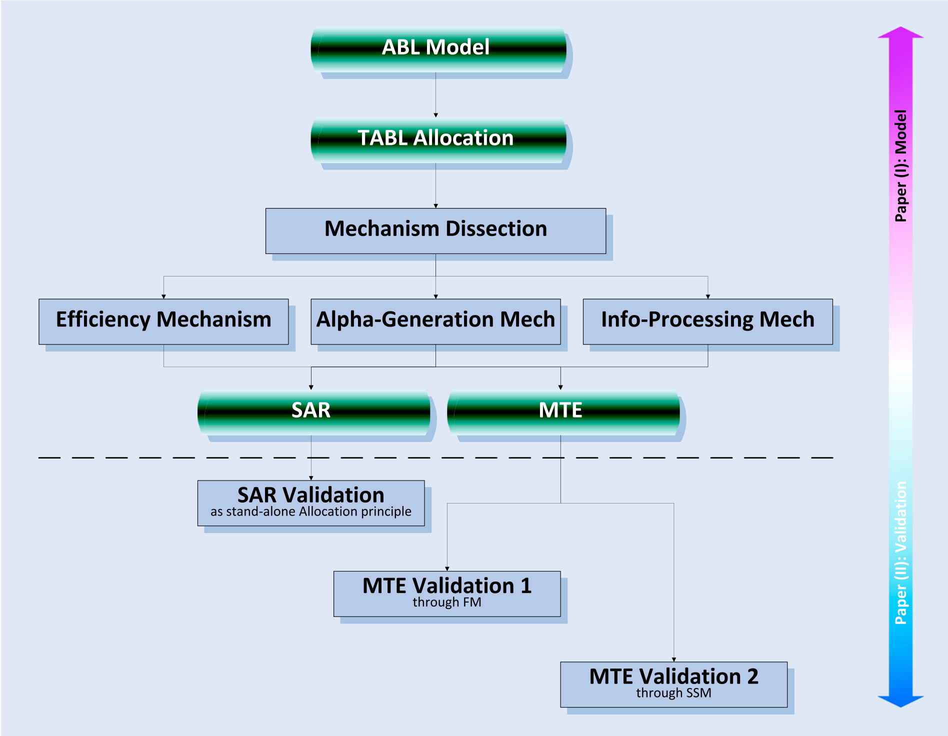

Figure 1. Research roadmap.

exemplifies this approach. Its mechanism dissection proceeds as follows:

(1) Derive the ABL allocation in closed form to visualise how heterogeneous investor behaviours are unified under a single equation (Section 3).
(2) Decompose the equation to hypothesise component functionalities and their combination mechanisms (Subsections 4.1.1 and 4.2.1).
(3) Extract the endogenous *mimicking* and *allocation* rules via symmetric pattern recognition, and derive their efficiency principles (Subsections 4.1.2 and 4.2.2).
(4) Investigate the *allocation* rule's alpha-generation and information-processing mechanisms, including ambiguous and subjective inputs (Subsections 4.3 and 4.4).
(5) Discuss whether these rules are *parsimonious* yet *complete* in forming a unified active PS framework (Subsection 4.5).
(6) Systematically validate these rules empirically as stand-alone concepts, and identify key performance drivers to ensure that the insights extend beyond ABL (Paper (II)).

Paper (I) covers steps 1–5; and Paper (II) addresses step 6.

While unconventional in finance, mechanism dissection suits this research. Portfolio theory has traditionally adhered to a *normative paradigm*, assuming perfect estimation and utility maximisation. Although research has progressively tackled estimation challenges in the MV framework† and refined utility representations to reflect human psychology‡, portfolio theories still struggle to coherently integrate *observed* investor PS behaviours (e.g. in Footnote 2) into a unified framework. In contrast, ABL's unification of diverse views (and hence behavioural patterns) makes dissection a natural tool to reveal core principles and inform a behaviourally grounded approach to active portfolio selection. By prioritising explanatory power over prescriptive formality, this paper resonates with the *descriptive* philosophy, while engaging with *normative* foundations.

Crucially, for mechanism dissection to be meaningful, the subject model's effectiveness must first be established; thus, highlighting ABL's merits is not promotional. The value of this research lies in exploring and validating parsimonious active PS principles and understanding mechanisms that coherently explain, integrate, and guide diverse investor behaviours. The following section solves the ABL model to derive a transparent allocation equation, enabling the dissection analysis.

3. The transparent ABL allocation model

Assume a market with n securities§ admits a linear factor model with f factors¶ :

$$\tilde{\boldsymbol{r}} = \boldsymbol{a} + \mathbf{B}\tilde{\boldsymbol{r}}_F + \tilde{\boldsymbol{\xi}} \tag{1}$$

† For example, Sharpe (1963), Cohen and Pogue (1967), Wallingford (1967), Rosenberg (1974), Alexander (1978), Markowitz and Perold (1981), Chan *et al.* (1999).

‡ For example, prospect theory (Kahneman and Tversky, 1979), Grishina *et al.* (2017), mental accounting theory (Thaler, 1985, Momen *et al.*, 2019), SP/A theory (Lopes, 1987, Shefrin and Statman, 2000), etc.

§ Such as tradable stocks, futures, swaps, bonds, etc.

¶ Such as Value, Quality, and Momentum, which influence asset pricing but are often not directly investable.

where $\widetilde{\boldsymbol{r}}$ is the $[n \times 1]$ vector of security excess returns (hereafter, 'return' refers to 'excess return', unless otherwise specified); $\boldsymbol{a}$ is the $[n \times 1]$ intercept vector; $\mathbf{B}$ is the $[n \times f]$ matrix of factor loadings or exposures; $\widetilde{\boldsymbol{r}}_F$ represents the $[f \times 1]$ vector of factor returns; and $\boldsymbol{\xi}$ stands for the $[n \times 1]$ vector of security-specific returns which are considered independent of each other and to the factor returns, following factor-modelling convention.

Its accompanying factor risk model is:

$$\boldsymbol{\Sigma} = \mathbf{B}\boldsymbol{\Sigma}_F\mathbf{B}^{\mathrm{T}} + \boldsymbol{\Sigma}_\xi \tag{2}$$

where $\boldsymbol{\Sigma}$ is the $[n \times n]$ covariance matrix of the security returns; $\boldsymbol{\Sigma}_F$ represents the $[f \times f]$ covariance matrix of the factor returns; and $\boldsymbol{\Sigma}_\xi$ stands for the $[n \times n]$ covariance matrix of the security-specific returns. $\boldsymbol{\Sigma}_\xi$ is diagonal due to independence.

In order to model how public ($\mathcal{G}$) and private ($\mathcal{H}$) information on securities, factors and security-specific returns propagate to the future performance of securities $\widetilde{\boldsymbol{r}}$, an augmented $[N \times 1]$ return vector $\widetilde{\boldsymbol{r}}^a$ (where $N = n + f$), incorporating both securities and factors, is considered:

$$\widetilde{\boldsymbol{r}}^a = \begin{pmatrix} \widetilde{\boldsymbol{r}} \\ \widetilde{\boldsymbol{r}}_F \end{pmatrix}. \tag{3}$$

Given linearity in (1), this augmentation also captures the security-specific information.

In accordance with the *semi-strong form of market efficiency* (Fama 1970), if only public information and techniques $\mathcal{G}$ are accessible, the augmented market view $\widetilde{\boldsymbol{\mu}}^a_{M[N\times 1]}$ should represent the best available view, superior to any individual's estimation $\widetilde{\boldsymbol{\mu}}^a_{[N\times 1]}$. Assuming the market is already in equilibrium, the $[n \times 1]$ security-level excess return vector can be assessed using the *Capital Asset Pricing Model* (CAPM), that is, $\widehat{\boldsymbol{\pi}} = \boldsymbol{\beta}[\mathbb{E}(\widetilde{R}_M \mid \mathcal{G}) - r_f]$, from the market portfolio $\boldsymbol{w}_{M[n\times 1]}$ where, $\boldsymbol{\beta}$ is the $[n \times 1]$ vector of security exposures to the market; $\mathbb{E}(\cdot)$ is the mathematical expectation operator; $\widetilde{R}_M$ is the long-term gross market return; $|\mathcal{G}$ means given public information; and r_f is the risk-free interest rate. The $[f \times 1]$ factor-level excess return vector, $\widehat{\boldsymbol{\pi}}_F$, can also be quantified.

However, the market is not necessarily in equilibrium. Therefore, the following normality assumption† regarding $\widetilde{\boldsymbol{\mu}}^a_M$ is made, which best represents $\widetilde{\boldsymbol{r}}^a$ given $\mathcal{G}$:

$$\widetilde{\boldsymbol{r}}^a_{|\mathcal{G}} \overset{\text{delegate}}{\rightarrow} \widetilde{\boldsymbol{\mu}}^a_M \sim \mathbb{N}(\widehat{\boldsymbol{\pi}}^a, \tau\boldsymbol{\Sigma}^a) \tag{4}$$

where $\mathbb{N}(\cdot)$ denotes the normal distribution;

$$\widehat{\boldsymbol{\pi}}^a = \begin{pmatrix} \widehat{\boldsymbol{\pi}} \\ \widehat{\boldsymbol{\pi}}_F \end{pmatrix} = \kappa \begin{pmatrix} \boldsymbol{\Sigma} \\ \boldsymbol{\Sigma}_F\mathbf{B}^{\mathrm{T}} \end{pmatrix} \boldsymbol{w}_M \tag{5}$$

is the augmented $[N \times 1]$ vector of CAPM-assessed equilibrium returns;

$$\kappa = \frac{\mathbb{E}(\widetilde{R}_M|\mathcal{G}) - r_f}{\sigma_M^2} \tag{6}$$

is the long-term market signal-noise ratio; σ_M is the long-term market risk;

$$\boldsymbol{\Sigma}^a = \begin{pmatrix} \boldsymbol{\Sigma} & \mathbf{B}\boldsymbol{\Sigma}_F \\ \boldsymbol{\Sigma}_F\mathbf{B}^{\mathrm{T}} & \boldsymbol{\Sigma}_F \end{pmatrix} \tag{7}$$

is the augmented $[N \times N]$ covariance matrix; and τ is a scalar representing the portfolio manager (PM)'s uncertainty about the covariance matrix estimation.

Expression (4) represents a *probabilistic view*, where the augmented market view $\widetilde{\boldsymbol{\mu}}^a_M$ is approximated by the equilibrium view $\widehat{\boldsymbol{\pi}}^a$, with error proportional to $\boldsymbol{\Sigma}^a$. When only public information is available, the *prior* view $\widetilde{\boldsymbol{r}}^a_{|\mathcal{G}}$ is best proxied by the market view $\widetilde{\boldsymbol{\mu}}^a_M$.

With private information $\mathcal{H}$, suppose the PM forms K ($\leqslant N$) views in the following format:

$$\mathbf{P}^a\widetilde{\boldsymbol{r}}^a_{|\mathcal{H},\mathcal{G}} = \widetilde{\boldsymbol{y}}^a_{|\mathcal{H},\mathcal{G}} + \widetilde{\boldsymbol{\varepsilon}}^a \tag{8}$$

where

$$\mathbf{P}^a = \begin{pmatrix} \mathbf{P} & \mathbf{0} \\ \mathbf{0} & \mathbf{P}_F \\ \mathbf{P}_\xi & -\mathbf{P}_\xi\mathbf{B} \end{pmatrix} \tag{9}$$

is the augmented $[K \times N]$ view structure matrix, containing k_1 ($\leqslant n$) security-level view structures (or strategies) $\mathbf{P}_{[k_1\times n]}$, k_2 ($\leqslant f$) factor-level view structures $\mathbf{P}_{F[k_2\times f]}$, k_3 ($\leqslant n$) security idiosyncratic level view structures $\mathbf{P}_{\xi[k_3\times n]}$, and $K = k_1 + k_2 + k_3$;

$$\widetilde{\boldsymbol{r}}^a_{|\mathcal{H},\mathcal{G}} = \begin{pmatrix} \widetilde{\boldsymbol{r}}_{|\mathcal{H},\mathcal{G}} \\ \widetilde{\boldsymbol{r}}_{F|\mathcal{H},\mathcal{G}} \end{pmatrix} \tag{10}$$

is the unknown but required, augmented $[N \times 1]$ *posterior* vector of returns, containing n posterior security returns $\widetilde{\boldsymbol{r}}_{|\mathcal{H},\mathcal{G}[n\times 1]}$, and f factor returns $\widetilde{\boldsymbol{r}}_{F|\mathcal{H},\mathcal{G}[f\times 1]}$, given information $\mathcal{G}$ and $\mathcal{H}$;

$$\widetilde{\boldsymbol{y}}^a_{|\mathcal{H},\mathcal{G}} = \begin{pmatrix} \widetilde{\boldsymbol{y}}_{|\mathcal{H},\mathcal{G}} \\ \widetilde{\boldsymbol{y}}_{F|\mathcal{H},\mathcal{G}} \\ \widetilde{\boldsymbol{y}}_{\xi|\mathcal{H},\mathcal{G}} \end{pmatrix} \tag{11}$$

is the augmented $[K \times 1]$ vector of updated views, containing k_1 views $\widetilde{\boldsymbol{y}}_{|\mathcal{H},\mathcal{G}[k_1\times 1]}$ on security returns, k_2 views $\widetilde{\boldsymbol{y}}_{F|\mathcal{H},\mathcal{G}[k_2\times 1]}$ on factor returns, k_3 views $\widetilde{\boldsymbol{y}}_{\xi|\mathcal{H},\mathcal{G}[k_3\times 1]}$ on security idiosyncratic returns; and

$$\widetilde{\boldsymbol{\varepsilon}}^a = \begin{pmatrix} \widetilde{\boldsymbol{\varepsilon}} \\ \widetilde{\boldsymbol{\varepsilon}}_F \\ \widetilde{\boldsymbol{\varepsilon}}_\xi \end{pmatrix} \tag{12}$$

is the augmented $[K \times 1]$ vector of view estimation errors, containing k_1 security-level view errors $\widetilde{\boldsymbol{\varepsilon}}_{[k_1\times 1]}$, k_2 factor-level

† Although ABL can handle *non-normal* markets with *non-linear* factor structures in Cheung (2009), this paper assumes *normality* for three reasons. First, it ensures the mathematical tractability needed for mechanism dissection. Second, evidence suggests that ignoring non-normality typically results in only small certainty-equivalent performance losses (Tu and Zhou 2004). Third, as Markowitz (2010) notes, starting with an MV framework does not preclude subsequent refinements for higher moments or non-normal features.

view errors $\widetilde{\boldsymbol{\varepsilon}}_{F[k_2\times 1]}$, k_3 security idiosyncratic level view errors $\widetilde{\boldsymbol{\varepsilon}}_{\xi[k_3\times 1]}$.

Formulation (8) allows a view to be expressed on a linear combination of securities, factors or security idiosyncratic returns, respectively. Suppose the error vector is normally distributed:

$$\widetilde{\boldsymbol{\varepsilon}}^a \sim \mathbb{N}(\mathbf{0}_{[K\times 1]}, \boldsymbol{\Omega}^a) \tag{13}$$

where $\mathbf{0}_{[K\times 1]}$ is the $[K \times 1]$ vector of zeros; and

$$\boldsymbol{\Omega}^a = \begin{pmatrix} \boldsymbol{\Omega} & & \mathbf{0} \\ & \boldsymbol{\Omega}_F & \\ \mathbf{0} & & \boldsymbol{\Omega}_\xi \end{pmatrix} \tag{14}$$

is the augmented $[K \times K]$ block-diagonal variance matrix of view-estimation errors (also referred to as *view noises*), provided by the PM, containing the security-level view uncertainty matrix $\boldsymbol{\Omega}_{[k_1\times k_1]}$, the factor-level view uncertainty matrix $\boldsymbol{\Omega}_{F[k_2\times k_2]}$, and the security idiosyncratic level view uncertainty matrix $\boldsymbol{\Omega}_{\xi[k_3\times k_3]}$. Note, just as a covariance matrix, the user-input view uncertainty matrix $\boldsymbol{\Omega}^a$ should be *real*, *symmetric* and *positive definite*.

By combining (8) and (13), the following is obtained:

$$\widetilde{\boldsymbol{y}}^a_{|\widetilde{\boldsymbol{r}}^a,\mathcal{H},\mathcal{G}} \sim \mathbb{N}(\mathbf{P}^a\widetilde{\boldsymbol{r}}^a_{|\mathcal{H},\mathcal{G}}, \boldsymbol{\Omega}^a). \tag{15}$$

These views provide valuable information for updating the outlook for $\widetilde{\boldsymbol{r}}^a$. Through the application of *Bayes' Rule* and the PM's 'ultimate' point estimates: $\widetilde{\boldsymbol{y}}^a_{|\widetilde{\boldsymbol{r}}^a,\mathcal{H},\mathcal{G}} \overset{\text{belief}}{\to} \widehat{\boldsymbol{q}}^a$, where

$$\widehat{\boldsymbol{q}}^a = \begin{pmatrix} \widehat{\boldsymbol{q}}_{[k_1\times 1]} \\ \widehat{\boldsymbol{q}}_{F[k_2\times 1]} \\ \widehat{\boldsymbol{q}}_{\xi[k_3\times 1]} \end{pmatrix} \tag{16}$$

is the augmented $[K \times 1]$ vector of best view point estimates (also referred to as *view signals*) provided by the PM, containing k_1 security-level view estimates $\widehat{\boldsymbol{q}}_{[k_1\times 1]}$, k_2 factor-level view estimates $\widehat{\boldsymbol{q}}_{F[k_2\times 1]}$, and k_3 security idiosyncratic level view estimates $\widehat{\boldsymbol{q}}_{\xi[k_3\times 1]}$, the ABL model is derived (Cheung,2013; Theorem A.1 in Appendix), yielding the *posterior* update for $\widetilde{\boldsymbol{r}}^a$.

The following model further solves the ABL allocation:

THEOREM 3.1 (The Transparent ABL (TABL) Allocation) *In the absence of constraints, the mean-variance efficient ABL posterior allocation admits the following representation:*

$$\boldsymbol{w}^* = \underbrace{\underbrace{\frac{\kappa}{2\lambda\tau}\boldsymbol{w}_M}_{\text{benchmark}} + \underbrace{\mathbf{P}^{\mathrm{T}}(2\lambda\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}}}_{\text{security/strategy combination tilt}}}_{\text{Original BL allocation}} + \underbrace{\underbrace{\mathbf{M}_{\mathrm{FM}}^{\mathrm{T}}\mathbf{P}_F^{\mathrm{T}}(2\lambda\boldsymbol{\Omega}_F)^{-1}\widehat{\boldsymbol{q}}_F}_{\text{composite factor-mimicking tilt}} + \underbrace{(\mathbf{I}_n - \mathbf{B}\mathbf{M}_{\mathrm{FM}})^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}(2\lambda\boldsymbol{\Omega}_\xi)^{-1}\widehat{\boldsymbol{q}}_\xi}_{\text{idiosyncratic tilt}}}_{\text{Augmented tilts in ABL}} \tag{17}$$

where $\boldsymbol{w}^*$ is the $[n \times 1]$ vector of optimal ABL allocation to the n securities; λ (> 0) is the PM's risk-aversion parameter; $\mathbf{I}_n$ is the $[n \times n]$ identity matrix;

$$\mathbf{M}_{\mathrm{FM}[f\times n]} = \left[(\boldsymbol{\Sigma}^+)^{-1}\mathbf{B}\boldsymbol{\Sigma}_F^+\right]^{\mathrm{T}}; \tag{18}$$

$\boldsymbol{\Sigma}^+_{[n\times n]} = \mathbf{B}\boldsymbol{\Sigma}_F^+\mathbf{B}^{\mathrm{T}} + \boldsymbol{\Sigma}_\xi^+$ *is the covariance matrix of the security returns updated by factor and security-specific views;* $\boldsymbol{\Sigma}^+_{F[f\times f]} = [(\tau\boldsymbol{\Sigma}_F)^{-1} + \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\mathbf{P}_F]^{-1}$ *is the covariance matrix of the factor returns updated by factor views; and* $\boldsymbol{\Sigma}^+_{\xi[n\times n]} = [(\tau\boldsymbol{\Sigma}_\xi)^{-1} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi]^{-1}$ *is the covariance matrix of the security-specific returns updated by security-specific views.*

Proof See Appendix. ■

REMARK Equation (17) provides a coherent (consistent, additive, and transparent) attribution of the optimal portfolio weights to investor views across benchmark, security, factor, and security-specific levels. This provides an explicit mathematical representation of ABL's unifying power over heterogeneous PS behaviours: *benchmark indexation*, *security/strategy combination tilt* (where '*tilt*' refers to active weights), *factor-mimicking tilt*, and *security-idiosyncratic tilt* (portable alpha).

Figure 2 illustrates the modular TABL allocation mechanism, demonstrating symmetry across view tilts. Subsequent examples in this paper and figures 5–8 of Paper (II) visualise active *security*, *basket*, and *security-specific* tilts. Cheung (2013) further illustrates factor-mimicking tilts.

Table 1 contrasts MPT, BL, ABL, and TABL by *modelling paradigm*, *investment style*, *view capacity*, and *behavioural compatibility*, etc. Note that while TABL is internally equivalent to ABL, it resolves ABL's opacity by providing structural transparency, thereby allowing the following section to dissect its efficiency principles and underlying economic mechanisms.

4. Deriving active portfolio-selection principles via dissection

4.1. *Mimicking principle*

4.1.1. Isolating the *Mimicking* and *Hedging* mechanisms.

From Theorem 3.1, it follows that:

HYPOTHESIS 4.1 (Factor Mimicking (FM)) $\mathbf{M}_{\mathrm{FM}}^{\mathrm{T}}$ *is the FM Transformation Matrix or Technique which converts a factor-level allocation into a security-level portfolio that replicates its performance (referred to as a FM tilt).*

REMARK Factor mimicking is essential when factors drive performance or risk but are not directly tradable (see e.g. Ferson *et al.* 2006). Suppose we wish to invest in or hedge against a non-tradable factor basket, or *composite factor*, $\boldsymbol{w}_{F[f\times 1]}$. ABL internally applies $\mathbf{M}_{\mathrm{FM}}^{\mathrm{T}}$ to transform $\boldsymbol{w}_F$ into a tradable security portfolio $\boldsymbol{w}^{[\mathrm{FM}]}_{[n\times 1]} = \mathbf{M}_{\mathrm{FM}}^{\mathrm{T}}\boldsymbol{w}_F$ that replicates its performance.

HYPOTHESIS 4.2 (Security-Specific Mimicking (SSM)) $(\mathbf{I} - \mathbf{B}\mathbf{M}_{\mathrm{FM}})^{\mathrm{T}}_{[n\times n]} = [(\boldsymbol{\Sigma}^+)^{-1}\boldsymbol{\Sigma}_\xi^+]$ *is the SSM Transformation*

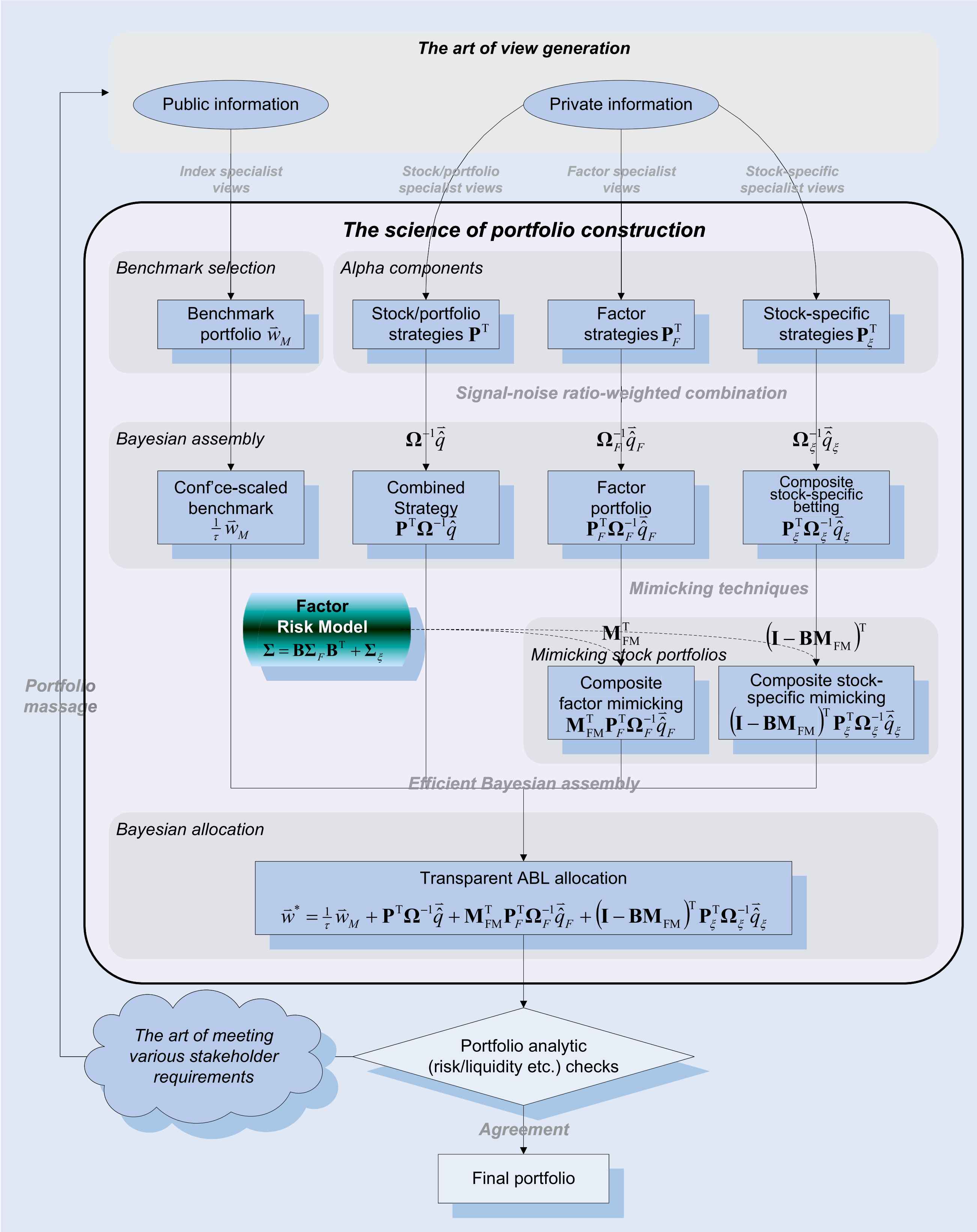

Figure 2. The transparent ABL allocation framework.
Note: Based on the subjectivity analysis in Subsection 4.4, it can be shown that the positive constants κ and 2λ can be omitted without loss of generality.

Table 1. Distinctive features of MPT, BL, ABL, and TABL.

Feature	MPT	BL	ABL	TABL
Modelling paradigm	Normative	Normative	Descriptive (view-only) & normative	Unifies descriptive & normative
Investment style	Passive	Unifies active & passive	Unifies active & passive	Unifies active & passive
View capacity	N/A	Security-level only	Unifies heterogeneous views*	Unifies heterogeneous views*
Behavioural compatibility	Disconnect	Opaque	Unifies behaviours in a black box	Unifies behaviours with transparency**
Relationship			ABL generalises BL	=ABL, but transparent
Source	Markowitz (1952 etc.)	Black and Litterman (1992)	Cheung (2013)	This paper

Note: * ABL and TABL support factor- and security-specific-level views, beyond BL's security-level views.** Unifying strategy combination, factor mimicking, hedging, portable alpha, etc., under a single equation.

Matrix or Technique which converts a security specific-level allocation into a security-level portfolio that replicates its performance (a SSM tilt or portable alpha).

Remark To gain exposure to a security basket $\boldsymbol{w}_{\xi[n\times1]}$ while hedging common factor risks, the model transforms $\boldsymbol{w}_\xi$ into a tradable security portfolio $\boldsymbol{w}^{[\mathrm{SSM}]}_{[n\times1]} = (\mathbf{I} - \mathbf{BM}_{\mathrm{FM}})^{\mathrm{T}}\boldsymbol{w}_\xi$ that replicates its idiosyncratic performance.

Hypothesis 4.3 (Hedging) $(\mathbf{BM}_{\mathrm{FM}})^{\mathrm{T}}_{[n\times n]}$ *is the Hedging Transformation Matrix or Technique that converts a security-level allocation into a portfolio that replicates its systematic performance, suitable for common risk hedging (a hedging portfolio).*

Remark To hedge the common risks of a security-level portfolio $\boldsymbol{w}_{\xi[n\times1]}$, the model uses $(\mathbf{BM}_{\mathrm{FM}})^{\mathrm{T}}_{[n\times n]}$ to generate a tradable portfolio $\boldsymbol{w}^{[\mathrm{H}]}_{[n\times1]} = (\mathbf{BM}_{\mathrm{FM}})^{\mathrm{T}}_{[n\times n]}\boldsymbol{w}_\xi$ that replicates its systematic performance.

4.1.2. Implied principle: minimum-tracking-error (MTE).

The following proposition confirms these hypotheses by revealing a unifying efficiency principle underlying the FM, SSM, and hedging techniques.

Proposition 4.1 (Principle 1: MTE-Guided Mimicking) *ABL inherently follows the Minimum-Tracking-Error (MTE) principle to guarantee efficient FM, SSM and hedging.*

Proof See Appendix. ∎

Remark The *MTE principle* refers to constructing a portfolio whose returns closely track a target market variable, minimising tracking variance to gain exposure to, or hedge against, the target.

The MTE-based FM technique aligns with Fama's (1996) multifactor-minimum-variance interpretation of Merton's (1973) ICAPM factor hedger. However, Proposition A.1 establishes MTE as ABL's endogenous efficiency principle rather than an imposed engineering choice. Cheung and Mittal (2009) demonstrate that this MTE-guided FM approach outperforms alternatives such as Fama-French ranking, ordinary least squares (OLS), and generalised least squares (GLS) in strengthening the correlations between factor tilts and their target factors.

4.2. Allocation principle

4.2.1. Isolating the *Allocation* mechanism.

From Theorem 3.1, a symmetric allocation mechanism emerges across security, factor, and security-specific levels of allocation: $\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}}$, $\boldsymbol{\Omega}_F^{-1}\widehat{\boldsymbol{q}}_F$, and $\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi$. Hence:

Hypothesis 4.4 (Subjective Allocation Rule (SAR)) *Across heterogeneous views, ABL combines and shrinks view strategies towards the benchmark based on their signal-noise (SN) ratios. For views $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Omega})$, where $\widehat{\boldsymbol{q}}$ is the signal vector and $\boldsymbol{\Omega}$ the noise matrix, the SN ratio vector is $\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}}$, yielding the SAR allocation:*

$$\boldsymbol{w}^* \propto \boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} \tag{19}$$

where $\boldsymbol{w}^*$ is the $[n\times1]$ tilt (active weight) vector; and $\propto$ denotes direct proportionality.

Remark The positive, subjective risk-aversion parameter λ is captured by the proportionality operator, which better reflects view ambiguity than strict equality ('$=$') as discussed in Subsection 4.4.2.

To see that the ABL allocation $\mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}}$ is the SN ratio-weighted combination of view strategies $\mathbf{P}$, consider the security (or portfolio)-level views specification (8):

$$\mathbf{P}\widetilde{\boldsymbol{r}} = \begin{pmatrix} \boldsymbol{p}_1^{\mathrm{T}} \\ \boldsymbol{p}_2^{\mathrm{T}} \\ \vdots \\ \boldsymbol{p}_{k_1}^{\mathrm{T}} \end{pmatrix} \widetilde{\boldsymbol{r}} = \widetilde{\boldsymbol{y}} + \widetilde{\boldsymbol{\varepsilon}} \quad \text{and} \quad \widetilde{\boldsymbol{\varepsilon}} \sim \mathbb{N}(\mathbf{0}_{[k_1\times1]}, \boldsymbol{\Omega}) \tag{20}$$

where $\boldsymbol{p}_j$ $(1 \leqslant j \leqslant k_1)$ is the j^{th} $[n\times1]$ vector of single-view structure vector, or the j^{th} *view strategy*; and $\mathbf{0}_{[k_1\times1]}$ is the $[k_1\times1]$ vector of zeros.

Table 2. Illustrative example of investor views.

Views	View structure $\mathbf{P}_{([3\times n])}^{T}$		
Securities	View 1	View 2	View 3
MMM	0	− 1	0.3
AOS	0	0	0
ABT	− 1	− 1	0
ACN	0	0	0.15
LULU	0	0	0
ADBE	0	3	0.3
AMD	1	0	0
AFL	0	− 1	0
…	…	…	…
ZION	0	0	0.25
View forecast $\widehat{\boldsymbol{q}}_{([1\times 3])}^{T}$	− 0.005	0.02	0.10
Uncertainty $\boldsymbol{\Omega}_{([3\times 3])}$	0.1	0	0
	0	0.2	0
	0	0	0.4

Suppose views are mutually independent; hence, $\boldsymbol{\Omega} = \mathrm{diag}\{\omega_1^2, \omega_2^2, \ldots, \omega_{k_1}^2\}$. Then $\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} = (\frac{\widehat{q}_1}{\omega_1^2}, \frac{\widehat{q}_2}{\omega_2^2}, \ldots, \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2})^{\mathrm{T}}$ gives the vector of view SN ratios, so that, $\mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} = \sum_{j=1}^{k_1} \frac{\widehat{q}_j}{\omega_j^2}\boldsymbol{p}_j$, confirming that view strategies are SN ratio-weighted. This generalises Elton and Gruber (2004) by using forward-looking views rather than backward-looking historical alpha-to-tracking variance ratios.

Table 2 illustrates the view structure matrix $\mathbf{P}^{\mathrm{T}}$, the mean vector $\widehat{\boldsymbol{q}}$, and the uncertainty matrix $\boldsymbol{\Omega}$ for three independent views: a pair-trading, a long/short, and a long-only strategy:

$$\begin{cases} -\widetilde{r}_{\mathrm{ABT}} + \widetilde{r}_{\mathrm{AMD}} \sim \mathbb{N}(-0.005, 0.1) \\ -\widetilde{r}_{\mathrm{MMM}} - \widetilde{r}_{\mathrm{ABT}} + 3\widetilde{r}_{\mathrm{ADBE}} - \widetilde{r}_{\mathrm{AFL}} \sim \mathbb{N}(0.02, 0.2) \\ 0.3\widetilde{r}_{\mathrm{MMM}} + 0.15\widetilde{r}_{\mathrm{ACN}} + 0.3\widetilde{r}_{\mathrm{ADBE}} + 0.25\widetilde{r}_{\mathrm{ZION}} \\ \quad \sim \mathbb{N}(0.10, 0.4) \end{cases} \tag{21}$$

The SAR tilt combines the three view strategies using SN ratio weighting like this:

$$\boldsymbol{w}^* \propto \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} = \frac{-0.005}{0.1}\begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ +1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{[n\times 1]} + \frac{0.02}{0.2}\begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 0 \\ +3 \\ 0 \\ -1 \\ \vdots \\ 0 \end{pmatrix}_{[n\times 1]} + \frac{0.10}{0.4}\begin{pmatrix} 0.3 \\ 0 \\ 0 \\ 0.15 \\ 0 \\ 0.3 \\ 0 \\ 0 \\ \vdots \\ 0.25 \end{pmatrix}_{[n\times 1]} \tag{22}$$

where $\boldsymbol{w}^*$ allocates active weights strictly according to the security sequence in table 1.

In general, for correlated views with a non-diagonal, symmetric, positive-definite view uncertainty matrix $\boldsymbol{\Omega}$, there exists an orthogonal matrix $\mathbf{Q}_{[k_1\times k_1]}$ (with property $\mathbf{Q}\mathbf{Q}^{\mathrm{T}} = \mathbf{I}_{k_1}$), such that

$$\boldsymbol{\Omega} = \mathbf{Q}\boldsymbol{\Omega}^{\perp}\mathbf{Q}^{\mathrm{T}} = \mathbf{Q}\begin{pmatrix} (\omega_1^{\perp})^2 & 0 & \cdots & 0 \\ 0 & (\omega_2^{\perp})^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & (\omega_{k_1}^{\perp})^2 \end{pmatrix}\mathbf{Q}^{\mathrm{T}} \tag{23}$$

where the diagonal entries $(\omega_j^{\perp})^2$ $(1 \leqslant j \leqslant k_1)$ are the eigenvalues of $\boldsymbol{\Omega}$.

Perform view space transform by $\mathbf{Q}^{\mathrm{T}}$, such that $\widetilde{\boldsymbol{y}} \to \widetilde{\boldsymbol{y}}_{[k_1\times 1]}^{\perp} = \mathbf{Q}^{\mathrm{T}}\widetilde{\boldsymbol{y}}$, $\widehat{\boldsymbol{q}} \to \widehat{\boldsymbol{q}}_{[k_1\times 1]}^{\perp} = \mathbf{Q}^{\mathrm{T}}\widehat{\boldsymbol{q}}$ and $\mathbf{P} \to \mathbf{P}_{[k_1\times n]}^{\perp} = \mathbf{Q}^{\mathrm{T}}\mathbf{P}$. In the new space, views become orthogonal: $\widetilde{\boldsymbol{y}}^{\perp} \sim \mathbb{N}(\widehat{\boldsymbol{q}}^{\perp}, \boldsymbol{\Omega}^{\perp})$. Noting $\boldsymbol{\Omega}^{-1} = \mathbf{Q}(\boldsymbol{\Omega}^{\perp})^{-1}\mathbf{Q}^{\mathrm{T}}$, the SAR tilt is then:

$$\mathbf{P}^{\mathrm{T}}(\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}} = (\mathbf{P}^{\perp})^{\mathrm{T}}\left(\boldsymbol{\Omega}^{\perp}\right)^{-1}\widehat{\boldsymbol{q}}^{\perp} = \sum_{j=1}^{k_1}\frac{\widehat{q}_j^{\perp}}{(\omega_j^{\perp})^2}\boldsymbol{p}_j^{\perp} \tag{24}$$

where $\boldsymbol{p}_{j[n\times 1]}^{\perp}$ $(1 \leqslant j \leqslant k_1)$ is the jth column vector of $(\mathbf{P}^{\perp})^{\mathrm{T}}$, that is, the jth view strategy in the orthogonal view space.

Thus, when views are correlated, ABL internally pursues the SN ratio-weighted view-mixing rule in the orthogonalised space. The same principle applies to factor and security-specific views, as ABL ultimately uses SAR to combine their respective mimicking portfolios.

4.2.2. Implied principle: Pareto efficiency (Diversification). Note SAR $\boldsymbol{\varpi}^* = (2\lambda\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}}$ solves the view-space signal-noise trade-off optimisation problem:

$$\boldsymbol{\varpi}^* = \underset{\boldsymbol{\varpi}}{\operatorname{argmax}}\left\{\boldsymbol{\varpi}^{\mathrm{T}}\widehat{\boldsymbol{q}} - \lambda\boldsymbol{\varpi}^{\mathrm{T}}\boldsymbol{\Omega}\boldsymbol{\varpi}\right\}, \tag{25}$$

where $\boldsymbol{\varpi}$ denotes the $[k_1 \times 1]$ view-level weight vector. Therefore, it follows:

LEMMA 4.1 (SAR Pareto Efficiency) *The SAR guarantees Pareto efficiency (i.e. diversification) with regard to view signals and noises.*

REMARK Lemma 4.1 implies that, through SN-efficient allocation, SAR intrinsically guarantees cross-view diversification. Note that, in this research, *Signal-noise (SN) efficiency* and *mean-variance (MV) efficiency* are both construed as Pareto efficiency since improving portfolio signal (or mean) inevitably increases portfolio noise (or variance), and *vice versa*.

4.2.3. Mindset shift: from *Objective* MV estimation to *Subjective* view expression.

By emphasising *SN efficiency* over *MV efficiency*, this research espouses a crucial shift in mindset. In the traditional MV approach, the *ex post* return distribution $\widetilde{\boldsymbol{r}}_{[n\times 1]} \sim \mathbb{N}(\widehat{\boldsymbol{\mu}}_{[n\times 1]}, \boldsymbol{\Sigma}_{[n\times n]})$, with mean vector $\widehat{\boldsymbol{\mu}}$ and covariance $\boldsymbol{\Sigma}$, differs fundamentally from the *ex ante* view distribution (15). The former relies on *historical* realised data, considered objective, whereas the latter captures *forward-looking*, uncertain, and subjective beliefs. For example, after a market decline, an *ex post* estimated model might be:

$$\begin{pmatrix}\widetilde{r}_A\\ \widetilde{r}_B\end{pmatrix} \sim \mathbb{N}\left[\begin{pmatrix}-0.1\\ -0.2\end{pmatrix}, \begin{pmatrix}0.09 & 0.048\\ 0.048 & 0.04\end{pmatrix}\right]. \tag{26}$$

Whereas if we instead anticipate stocks A and B to rebound by 10% and 20%, respectively, with stronger conviction in A and assuming independent views, the following views may be formed:

$$\begin{pmatrix}\widetilde{r}_A\\ \widetilde{r}_B\end{pmatrix} \sim \mathbb{N}\left[\begin{pmatrix}0.1\\ 0.2\end{pmatrix}, \begin{pmatrix}0.04 & 0\\ 0 & 0.16\end{pmatrix}\right], \tag{27}$$

where the signal vector $(0.1, 0.2)^{\mathrm{T}}$ specifies the forecasts, and, due to view independence, the noise matrix is diagonal: diag$(0.04, 0.16)$, capturing view uncertainties, with lower values indicating higher conviction. These *ex ante* views are purely subjective and may not align with the market model (26) at all.

Traditional MV optimisation is exacting: it requires full estimation of the mean vector $\widehat{\boldsymbol{\mu}}$ and the covariance matrix $\boldsymbol{\Sigma}$, and even a single missing element or minor errors can destabilise the optimiser, causing breakdowns or high turnover. These rigid requirements are especially challenging in high-dimensional settings. In contrast, ABL's subjective approach frees investors from estimating the full universe, allowing them to infer from a few user-selected familiar factors (e.g. Value, Brexit). This reduces dimensionality and transforms portfolio selection from traditional *objective*, abstract 'rocket science' to *subjective*, intuitive 'portfolio discovery'.

To signify this crucial mindset shift, this paper prioritises *SN efficiency* over *MV efficiency*. The shift accommodates ambiguous information (Subsection 4.4), responds to Treynor and Black (1973)'s call to reconcile subjective views with objective selection, and addresses the seven challenges identified by Salo *et al.* (2024) (Section 6). Regarding its economic value, Paper (II) develops specialised techniques to evaluate how subjectivity affects performance.

4.3. *SAR value-adding mechanism: information ratio proportional to skill*

To see how SAR adds value, replace the point estimates $\widehat{\boldsymbol{q}}, \widehat{\boldsymbol{q}}_F$, and $\widehat{\boldsymbol{q}}_\xi$ with the probabilistic views $\widetilde{\boldsymbol{y}}, \widetilde{\boldsymbol{y}}_F$, and $\widetilde{\boldsymbol{y}}_\xi$, respectively, which represent predicted (*ex ante*) view strategy returns. The expected view-driven tilt alpha (outperformance) is then:

$$\begin{aligned}
&\mathbb{E}\left[(\boldsymbol{w}^*)^{\mathrm{T}}\widetilde{\boldsymbol{r}}\right]\\
&\quad= \mathbb{E}\Big[\widetilde{\boldsymbol{y}}^{\mathrm{T}}(2\lambda\boldsymbol{\Omega})^{-1}\mathbf{P}\widetilde{\boldsymbol{r}} + \widetilde{\boldsymbol{y}}_F^{\mathrm{T}}(2\lambda\boldsymbol{\Omega}_F)^{-1}\mathbf{P}_F\mathbf{M}_{\mathrm{FM}}\widetilde{\boldsymbol{r}}\\
&\qquad + \widetilde{\boldsymbol{y}}_\xi^{\mathrm{T}}\left(2\lambda\boldsymbol{\Omega}_\xi\right)^{-1}\mathbf{P}_\xi\left(\mathbf{I}-\mathbf{B}\mathbf{M}_{\mathrm{FM}}\right)\widetilde{\boldsymbol{r}}\Big]\\
&\quad\underset{(A25)\&(A26)}{\overset{\mathrm{MTE}}{=}} \mathbb{E}\left[\widetilde{\boldsymbol{y}}^{\mathrm{T}}(2\lambda\boldsymbol{\Omega})^{-1}\mathbf{P}\widetilde{\boldsymbol{r}}\right] + \mathbb{E}\left[\widetilde{\boldsymbol{y}}_F^{\mathrm{T}}(2\lambda\boldsymbol{\Omega}_F)^{-1}\mathbf{P}_F\widetilde{\boldsymbol{r}}_F\right]\\
&\qquad + \mathbb{E}\left[\widetilde{\boldsymbol{y}}_\xi^{\mathrm{T}}\left(2\lambda\boldsymbol{\Omega}_\xi\right)^{-1}\mathbf{P}_\xi\widetilde{\boldsymbol{\xi}}\right]\\
&\quad= \mathbb{E}\left[\widetilde{\boldsymbol{y}}^{\mathrm{T}}(2\lambda\boldsymbol{\Omega})^{-1}\widetilde{\boldsymbol{x}}\right] + \mathbb{E}\left[\widetilde{\boldsymbol{y}}_F^{\mathrm{T}}(2\lambda\boldsymbol{\Omega}_F)^{-1}\widetilde{\boldsymbol{x}}_F\right]\\
&\qquad + \mathbb{E}\left[\widetilde{\boldsymbol{y}}_\xi^{\mathrm{T}}\left(2\lambda\boldsymbol{\Omega}_\xi\right)^{-1}\widetilde{\boldsymbol{x}}_\xi\right]
\end{aligned} \tag{28}$$

where the second equality uses the performance-mimicking results as explained in (A25) and (A26) in Appendix; $\widetilde{\boldsymbol{x}}, \widetilde{\boldsymbol{x}}_F$, and $\widetilde{\boldsymbol{x}}_\xi$ denote the $[k_1 \times 1]$, $[k_2 \times 1]$, and $[k_3 \times 1]$ vectors of realised (*ex post*) view strategy returns for securities (or portfolios) $\mathbf{P}\widetilde{\boldsymbol{r}}$, factors (or composite factors) $\mathbf{P}_F\widetilde{\boldsymbol{r}}_F$, and security-specific views (or their composites) $\mathbf{P}_\xi\widetilde{\boldsymbol{\xi}}$, respectively.

Suppose analysts are skillful, with all views independently and directionally correct ($\widetilde{\boldsymbol{x}} \propto \widetilde{\boldsymbol{y}}, \widetilde{\boldsymbol{x}}_F \propto \widetilde{\boldsymbol{y}}_F$ and $\widetilde{\boldsymbol{x}}_\xi \propto \widetilde{\boldsymbol{y}}_\xi$). Since $(2\lambda\boldsymbol{\Omega})^{-1}$, $(2\lambda\boldsymbol{\Omega}_F)^{-1}$, and $(2\lambda\boldsymbol{\Omega}_\xi)^{-1}$ are positive definite, the tilt alphas are positive-definite quadratic forms. Thus, SAR guarantees positive alpha generation.

In general, SAR guarantees risk-adjusted alpha proportional to forecasting skill:

LEMMA 4.2 (SAR Alpha Generation with Orthogonal Views) *SAR generates an information ratio (IR) directly proportional to view skill (IC) and the betting scale. When views are both skilled and mutually orthogonal, SAR guarantees aggregate alpha, without compounding errors.*

Proof See Appendix. ■

REMARK (1) *Forecasting skill*, or *information coefficient* (IC), is the correlation between historical predicted and realised returns (Jensen 1972). (2) Grinold (1994) introduces the rule of thumb that 'alpha is volatility times IC times score'. Our first-principles derivation (A30) yields

$$\mathrm{IR} = \mathrm{IC} \times \text{betting scale}, \tag{29}$$

which rigorously concretises the 'score' as a *betting scale* (> 0). Hence, IR is directly proportional to IC and to the betting scale. (3) As shown in (A29), with orthogonal views, errors from individual views, mimicking, or hedging are *additive*, not *multiplicative*; for many independent views, the Law of Large Numbers ensures aggregate errors diminish (Ross 1976, p. 342), thereby *diversifying*, rather than *compounding*. (4) Subsection 4.4.3 demonstrates that even when orthogonality is relaxed, SAR still extracts alpha from positively skilled views without compounding errors.

The Joint-Test Issue in Portfolio Theory Validation. Lemma 4.2 implies that *SAR serves as an efficient conduit, translating investor predictions into portfolio outcomes rather than generating alpha itself*. This principle applies to all PS models, especially in modern asset management where analysts and alpha models focus on forecasting. Consequently, PS model evaluations should prioritise skill-to-performance conversion efficiency rather than raw profitability. Assessing models via profitability 'horse races' confounds model efficiency with input quality, producing inconclusive *joint tests*: sound models may underperform with poor inputs, while strong predictions can make inefficient models appear effective. Accurate evaluation thus requires separating model efficiency from input quality – a methodology developed in Paper (II).

4.4. SAR cognition-friendly mechanism: portfolio massage

4.4.1. Descriptive characterisation of view ambiguity or subjectivity. While the MV approach demands high estimation accuracy, investor views are typically expressed *directionally* rather than *quantitatively* – for example, 'a 3-month bullish outlook' instead of 'a 20.5% rally in 3 months'; analyst ratings likewise follow an *ordinal* scale (Buy > Hold > Sell). This suggests that practitioners are more attuned to view *direction* than *strength*.

The traditional MV framework, being *normative*, embeds ambiguity or subjectivity into either the utility function (behavioural economics; see Footnote 12) or the return distribution (robust optimisation; e.g. Goldfarb and Iyengar 2003; Anderson and Cheng 2016; Yin *et al.* 2021, Blanchet *et al.* 2022). Despite their rigour, these methods are often too rigid to capture the nuances of real-world PS behaviours.

To bridge theory and practice, this paper develops a **descriptive characterisation methodology**, designed to explicitly capture view ambiguity or subjectivity through:

ASSUMPTION 4.1 (Strength-Agnostic Directional Views) *Upon receiving information $\widetilde{r} \sim \mathbb{N}(\widehat{q}, \omega^2)$ (where $\widetilde{r}$ is the return outlook; $\widehat{q}$ the signal; and ω^2 the noise), investors are view strength-agnostic, perceiving all views in the family $\{\widetilde{r} \sim \mathbb{N}(\widehat{q}, \eta\omega^2) \mid \forall \eta > 0\}$ as cognitively indistinguishable or subjectively equivalent in direction.*

REMARK In practice, the signal $\widehat{q}$ may appear as a full return *forecast*, a ranking *score*, or even a *directional signal* $\mathrm{sign}(\widehat{q})$ (where $\mathrm{sign}(\cdot)$ maps negatives to -1, positives to $+1$, and leaves zeros unchanged), capturing analyst ratings, practitioner sign-based investing behaviour (Buckle 2007), and generalising inequality-based views (Chiarawongse *et al.* 2012). Despite discarding magnitude, the Bussgang theorem implies that for Gaussian signals the *directional signal* retains approximately $\sqrt{2/\pi} \approx 0.7979$ of the original skill. The portfolio implications are examined in Paper (II). Although η represents view strength, investors typically omit specifying it when forming views.

Views are inherently uncertain, and quantifying cross-view correlations is even more so. Analysts tend to formulate forecasts in isolation, and investors frequently adopt them as independent – a sign of cognitive detachment or subjective indifference to correlation, as captured by:

ASSUMPTION 4.2 (Correlation-Agnostic Orthogonal Views) *Upon receiving k views $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Omega})$ (where $\widetilde{\boldsymbol{r}}$ is the $[k \times 1]$ vector of return outlooks; $\widehat{\boldsymbol{q}}$ the $[k \times 1]$ signal vector; $\boldsymbol{\Omega}$ the $[k \times k]$ noise matrix with non-zero off-diagonals), investors, being correlation-agnostic, treat the views as orthogonal: $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Lambda})$ (where $\boldsymbol{\Lambda}$ is a $[k \times k]$ positive-definite diagonal proxy for $\boldsymbol{\Omega}$).*

REMARK Consistent with Assumption 4.1, investors may freely assign positive values to the diagonal of $\boldsymbol{\Lambda}$ when forming views, so $\boldsymbol{\Lambda}$ need not correspond to $\boldsymbol{\Omega}$.

Together, the two assumptions capture real-world investor information-processing behaviour through **orthogonal directional views**. The remainder of this subsection examines how SAR effectively harnesses such ambiguous and subjective views to guide portfolio selection.

4.4.2. Portfolio massage: Operationalising orthogonal directional views.

LEMMA 4.3 (SAR Maps Indistinguishable Views to Distinct Portfolios) *SAR maps indistinguishable orthogonal directional views of varying strengths into analytically distinct portfolios.*

Proof See Appendix. ■

Investors are cognitively indifferent between k orthogonal directional views with varying uncertainties, provided their directions are preserved (Assumption 4.1). However, SAR maps these *indistinguishable* views to *analytically distinct* portfolios (Lemma 4.3), each carrying welfare implications that investors can discern. This enables investors to treat view uncertainties as k degrees of freedom to enhance welfare (analytical exposures), resulting in *Pareto improvements* – achieving *strategic* exposure goals without compromising *tactical* view implementation – distinct from *Pareto efficiency* in Lemma 4.1. Such operations are termed:

DEFINITION 4.1 (Portfolio Massage) *Portfolio massage allows investors to adjust view uncertainties (the diagonal elements of $\boldsymbol{\Lambda}$) to better align portfolios with their strategic exposure goals, with views added or removed as needed.*

Note that portfolio massage fully operationalises Assumptions 4.1 and 4.2:

LEMMA 4.4 (Representation of Orthogonal Directional Views) *Portfolio massage sufficiently operationalises orthogonal directional views.*

Proof See Appendix. ■

Figure 3 illustrates how analytics-guided portfolio massage calibrates view uncertainties. For example, with views in (27), SAR allocation is $\boldsymbol{w}^* \propto \boldsymbol{\Lambda}^{-1}\boldsymbol{q} = \begin{pmatrix}\frac{0.1}{0.04} & \frac{0.2}{0.16}\end{pmatrix}^{\mathrm{T}} = \begin{pmatrix}2.50 & 1.25\end{pmatrix}^{\mathrm{T}}$. To scale the 375% total weight to 100%, the traditional *rocket-science* approach imposes a *fully-invested* optimiser constraint, potentially distorting views. *Portfolio massage* instead homogenously scales uncertainties by 3.75, yielding a moderate-conviction portfolio $\begin{pmatrix}66.7\% & 33.3\%\end{pmatrix}^{\mathrm{T}}$, preserving view directions while handling ambiguity via the proportion operator '$\propto$'.

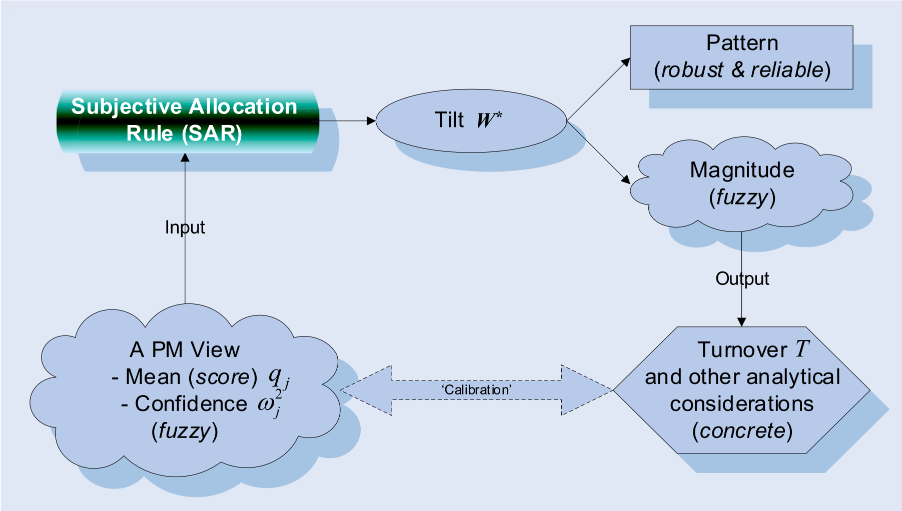

Figure 3. Analytics-guided portfolio massage.
Notes: This figure illustrates the concept of SAR-based, analytics-guided portfolio massage: (1) Investors input a fuzzy directional view with a signal score q_j and a tentative noise ω_j^2. (2) SAR map the view to a tilt $\boldsymbol{w}^*$, whose pattern is reliable, but magnitude is fuzzy. (3) Analytics reports the tilt's turnover, factor exposures etc. (4) Judging from the analytical feedback, investors adjust their view strength and, if needed, add new views to reflect their true intentions. This process continues until a portfolio is reached that balances tactical needs and strategic goals.

If equal weighting better aligns with the investor's strategic goal, adjusting the two uncertainties to 0.20 and 0.40,

$$\begin{pmatrix} \widetilde{r}_A \\ \widetilde{r}_B \end{pmatrix} \sim \mathbb{N}\left[\begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix}, \begin{pmatrix} 0.20 & 0 \\ 0 & 0.40 \end{pmatrix}\right], \tag{30}$$

yields a 50/50 allocation, enhancing portfolio welfare analytically while preserving view integrity directionally (Assumption 4.1).

Portfolio massage incorporates additional information, such as turnover, weight, or factor exposure targets, to calibrate views. Once a balanced portfolio is achieved, ambiguity is fully resolved, as in (30), with the views readily implemented in the portfolio. Using a realistic worked example, Section 5 demonstrates portfolio massage in practice, step by step.

4.4.3. Correlated views can always be equivalently represented by orthogonal views.

While SAR generates alpha given orthogonal views (Lemma 4.2), could correlated views impair its alpha-generating effectiveness? Suppose $\widetilde{r}_1$ and $\widetilde{r}_2$ are identical (e.g. HSBA LN vs. 5 HK, or SPY vs. VOO), while $\widetilde{r}_3$ is independent. Being correlation-unaware, we treat all three views as orthogonal (Assumption 4.2):

$$\begin{pmatrix} \widetilde{r}_1 \\ \widetilde{r}_2 \\ \widetilde{r}_3 \end{pmatrix} \sim \mathbb{N}\left[\begin{pmatrix} \widehat{q}_1 \\ \widehat{q}_2 \\ \widehat{q}_3 \end{pmatrix}, \begin{pmatrix} \omega_1^2 & 0 & 0 \\ 0 & \omega_2^2 & 0 \\ 0 & 0 & \omega_3^2 \end{pmatrix}\right]. \tag{31}$$

The SAR tilt vector is:

$$\boldsymbol{w}^* \propto \begin{pmatrix} \dfrac{\widehat{q}_1}{\omega_1^2} \\ \dfrac{\widehat{q}_2}{\omega_2^2} \\ \dfrac{\widehat{q}_3}{\omega_3^2} \end{pmatrix}. \tag{32}$$

Its return is:

$$\begin{aligned} (\boldsymbol{w}^*)^{\mathrm{T}} \begin{pmatrix} \widetilde{r}_1 \\ \widetilde{r}_2 \\ \widetilde{r}_3 \end{pmatrix} &\propto \frac{\widehat{q}_1}{\omega_1^2}\widetilde{r}_1 + \frac{\widehat{q}_2}{\omega_2^2}\widetilde{r}_2 + \frac{\widehat{q}_3}{\omega_3^2}\widetilde{r}_3 \\ &= \left(\frac{\widehat{q}_1}{\omega_1^2} + \frac{\widehat{q}_2}{\omega_2^2}\right)\widetilde{r}_1 + \frac{\widehat{q}_3}{\omega_3^2}\widetilde{r}_3. \end{aligned} \tag{33}$$

Since $\widetilde{r}_1 = \widetilde{r}_2$, allocations to the same factor, security 1 (or 2), become large. While these views may generate alpha, the associated risk stems from this concentration, so a crash in the factor could cause significant losses.

However, this risk can be readily detected using portfolio risk-attribution analytics:

$$\mathbb{V}\mathrm{ar}\left[(\boldsymbol{w}^*)^{\mathrm{T}} \begin{pmatrix} \widetilde{r}_1 \\ \widetilde{r}_2 \\ \widetilde{r}_3 \end{pmatrix}\right] \propto \left(\frac{\widehat{q}_1}{\omega_1^2} + \frac{\widehat{q}_2}{\omega_2^2}\right)^2 \sigma_1^2 + \left(\frac{\widehat{q}_3}{\omega_3^2}\right)^2 \sigma_3^2. \tag{34}$$

Noticing heavy risk exposure $(\frac{\widehat{q}_1}{\omega_1^2} + \frac{\widehat{q}_2}{\omega_2^2})^2\sigma_1^2$ to the factor $\widetilde{r}_1$ (or $\widetilde{r}_2$), we can massage, say, view 2 by setting ω_2^2 very large, effectively abandoning the view. Thus, even correlation-unaware, analytics-guided portfolio massage can help mitigate issues from high correlations.

In general, SAR preserves its effectiveness even in the presence of correlated views because:

Lemma 4.5 (Orthogonal Representation of Correlated Views) *Under SAR, correlated views can always be represented welfare-equivalently by orthogonal views through portfolio massage.*

Proof See Appendix. ■

Remark The welfare-equivalent orthogonal representation (A.32) is inherent. While it can be attained via analytics-guided portfolio massage (see figure 4 for a two-view geometric intuition), it needs not be derived explicitly in practice; instead, it suffices to follow slight 'goalkeeping' guidance at the end of the proof to avoid incorporating excessively correlated, mutually value-damaging views.

This is good news for investors who are truly agnostic about correlations (Assumption 4.2), since all what matters to them are portfolio exposures, which determine *welfare*, whether or not the original *subjective* and *ambiguous* views are represented precisely.

4.4.4. Resolving ambiguity with analytical information: Pareto improvements. Thus, rather than striving for unrealistic estimation accuracy, investors should prioritise achieving an analytically validated portfolio that reflects their true investment rationale – balancing *tactical* views with *strategic* exposures. The following establishes the feasibility of this approach:

Proposition 4.2 (Feasibility of Pareto Exposure Improvements) *Under SAR, portfolio massage leverages additional information to quantify and implement ambiguous views, converting ambiguity into Pareto improvements aligned with strategic goals while preserving tactical views. Improvements can be manual, but well-defined exposure objectives enable optimisation.*

Proof See Appendix. ■

Remark Achieving ideal exposures (welfare) through analytics-guided portfolio massage naturally resolves view ambiguities. Although one-shot optimisation is possible, rationality-bounded investors may find *view-driven, feedback-responsive* portfolio discovery more cognition-aligned, as it helps them uncover their underlying investment rationale (see the worked example in Section 5). This *view-driven, feedback-responsive* process is akin to backward induction as implemented in AI, albeit in a small-data setting.

4.4.5. Portfolio massage and existing approaches to ambiguity and subjectivity. Portfolio massage complements existing research on ambiguity and subjectivity.

Normative and Descriptive Decision Theory. Modelling portfolio selection as a discovery process aligns with Newell and Simon's (1956) symbolic modelling method, which views decision-making as heuristic-guided search under cognitive constraints – a foundation for AI, cognitive science, and *descriptive decision theory*. Similarly, portfolio massage iteratively uncovers the investor's true rationale, reflecting a rational but cognitively bounded decision-maker navigating factor exposures and opinion trade-offs. While traditional portfolio theory is primarily *normative*, this paper bridges the *normative* and *descriptive* paradigms, better capturing dynamic utility (Simon 2000) under uncertainty and providing a richer view of real-world behaviours.

Robust Optimisation. Unlike robust optimisation, which seeks *conservative* solutions under uncertainty, portfolio massage embraces ambiguity to *actively* pursue Pareto improvements, aligning portfolio exposures with *strategic* goals without compromising *tactical* view implementation.

Subjectivity and Ambiguity Discussions in Finance. While the literature explores subjectivity representation (e.g. Smimou *et al.* 2008), portfolio optimisation under ambiguity (e.g. Pflug and Wozabal 2007, Hasuike and Katagiri 2010, Boyle *et al.* 2012), endogenised information choices (e.g. Van Nieuwerburgh and Veldkamp 2010), and investor perception surveys (e.g. Calvo-Pardo *et al.* 2022), simple, practical techniques that are closely connected to real-world behaviour remain underexplored. SAR-based portfolio massage integrates qualitative and quantitative information without adding complexity, providing a natural, scalable way to address subjectivity and ambiguity.

4.4.6. SAR general alpha generation. Taken together, it follows:

Proposition 4.3 (Principle 2: SAR-Based Allocation) *With analytics-guided portfolio massage, the signal-noise (SN) ratio-weighted SAR can Pareto-efficiently generate alpha proportional to view skills without compounding errors.*

Proof SAR guarantees alpha generation from orthogonal, skilled views without compounding errors (Lemma 4.2) while preserving cross-view diversification (Lemma 4.1). Through portfolio massage, even correlated views can be treated welfare-equivalently as orthogonal (Lemma 4.5). Hence, as long as all views – independent or correlated, quantitative or qualitative, objective or subjective – exhibit positive skill, SAR ensures outperformance. ■

Remark SAR is a versatile and efficient conduit for translating diverse forecasting skills into alpha.

4.5. Principles of active portfolio selection

4.5.1. SAR and MTE span a comprehensive active portfolio-selection framework. Since MTE allows the broadest views, including factor and idiosyncratic views (Proposition 4.1), to be efficiently mapped to portfolios, and SAR ensures that all positive-skilled views are combined in a Pareto-efficient (i.e. diversified) and alpha-generating manner (Proposition 4.3), it follows:

Theorem 4.1 (View-to-Portfolio Mapping) *SAR-based allocation and MTE-guided mimicking and hedging, whether modular or combined, guarantee efficient, alpha-generating view-to-portfolio mapping without compounding errors. Together, they span a comprehensive active portfolio-selection framework.*

Remark Like two independent vectors spanning a plane, SAR and MTE together form a versatile *Bayesian Allocation Framework*, emcompassing a continuum of efficient PS solutions, including – and extending beyond – ABL. This framework enables investors to harness the broadest

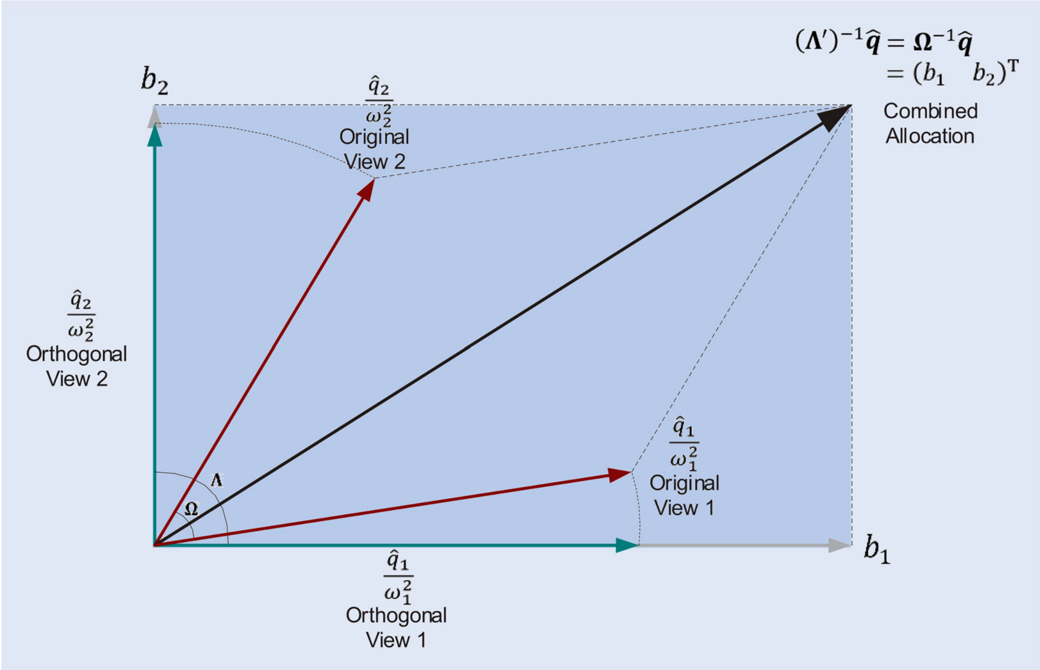

Figure 4. Welfare-equivalent orthogonal representation of correlated views.
Notes: This figure illustrates the geometric intuition of the equivalent orthogonal representation of correlated views. While the original views $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Omega})$ (where $\widehat{\boldsymbol{q}} = (\widehat{q}_1\ \widehat{q}_2)^{\mathrm{T}}$) may be correlated, correlation-agnostic investors process them as orthogonal views $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Lambda})$. A diagonal matrix $\boldsymbol{\Lambda}'$ (with diagonal elements $\frac{\widehat{q}_1}{b_1}$ and $\frac{\widehat{q}_2}{b_2}$), reachable through portfolio massage, ensures the SAR allocation $(\boldsymbol{\Lambda}')^{-1}\widehat{\boldsymbol{q}}$ matches $(\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}}$. Given the subjective and fuzzy nature of views and the importance of portfolios for investor welfare, using the orthogonal views $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Lambda}')$ to represent the original correlated views $\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Omega})$ is welfare-equivalent.

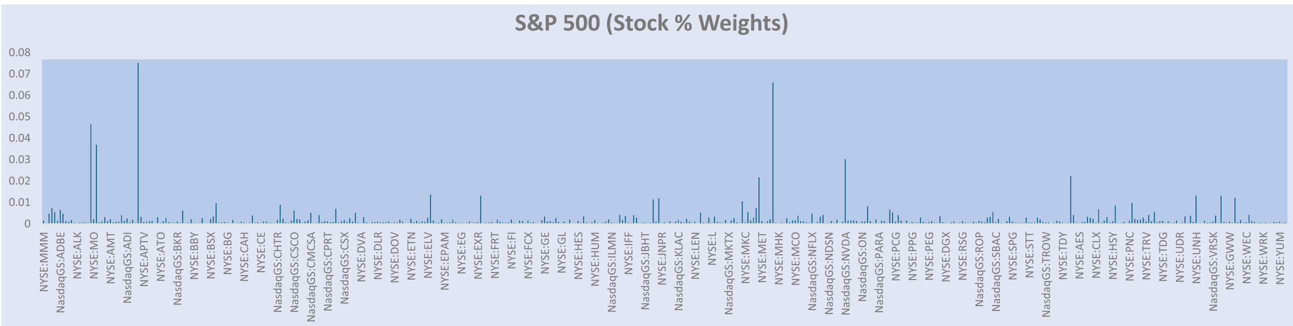

Figure 5. The initial indexation portfolio.

information sources while ensuring their efficient, alpha-generating implementation. Even within the same market setting, the framework refracts into distinct optimal strategies, reflecting investors' cognition, forecasting skill, and subjectivity.

4.5.2. Parsimony and completeness. Given SAR and MTE respectively govern distinct and necessary *allocation* and *mimicking* functions, and that these two principles sufficiently span the Bayesian Allocation Framework (Theorem 4.1) – unifying at least *strategy combination*, *equal weighting*,† *factor mimicking*, *hedging*, and *portable alpha* – it follows:

Corollary 4.1 (Principles of Active Portfolio Selection) *If portfolio-selection practices can be generally decomposed into allocation and mimicking sub-problems, SAR and MTE constitute the parsimonious and complete principles of active portfolio selection.*

Remark This statement assumes the existence of a general decomposition of PS practices. Forthcoming research (Cheung 2026) will rigorously test this assumption, thereby completing the logical cycle to demonstrate the completeness of SAR and MTE.

5. A worked example: holistic factor management

Among the framework's many applications, this section illustrates *holistic factor management* (HFM; Cheung and Mishra 2010b), a long-standing challenge in modern portfolio management. Using realistic data, it illustrates how *portfolio massage*, SAR, and MTE enable intuitive, *view-driven*, and

† Equal weighting, though not explicit in Theorem 3.1, is a special case of SAR as in (30).

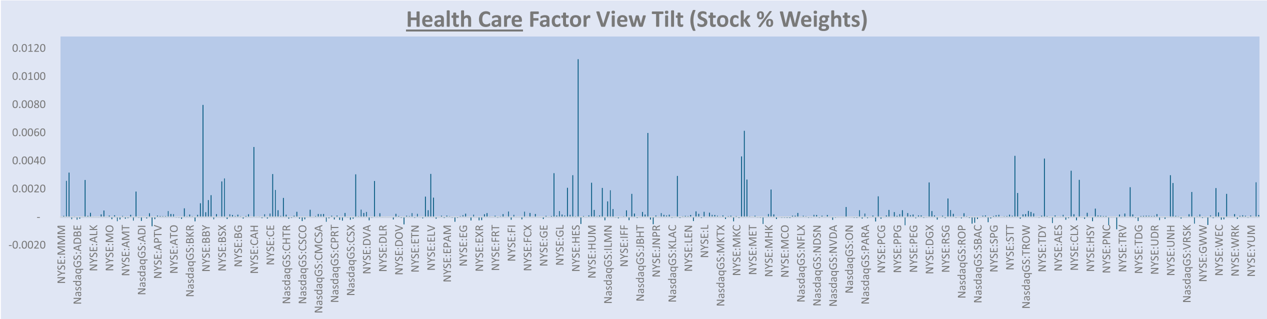

Figure 6. The active weights of the initial single factor view.

feedback-responsive discovery of the portfolio with ideal exposures.

As fund size grows, shifting from security-level monitoring to HFM reduces dimensionality while preserving tactical and strategic intent. For example, within the S&P 500 universe, a fund manager who *strategically* favours high *Dividend Yield* ('DY') stocks may *tactically* enhance their indexation strategy (Figure 5) based on a bullish view on the *Health Care* ('HCare') factor. In this framework, analytics-guided portfolio massage helps them discover their ideal portfolio:

Step 1 *View-to-Portfolio Mapping*. The manager submits a bullish factor-level view $\widetilde{r}_{\text{HCare}} \sim \mathbb{N}(0.1, 0.1)$, representing a directional belief with arbitrary signal and uncertainty magnitudes. MTE then translates this view into an initial healthcare FM tilt (Figure 6).

Step 2 *Analytic Feedback*. Examining the tilt (active weights) (Figure 6) and its factor exposures (Figure 7), the investor recognises the expected concentration of active weights in *Health Care* stocks, the satisfactory magnitude of tilts, and the intended, sharp HCare factor exposure. The minimal exposures to other factors are appropriate, except for the *negative* exposure to DY, which contradicts their strategic view.

Step 3 *View-to-Portfolio Mapping*. To reflect their *strategic* bullish stance on DY, an additional bullish DY view $\widetilde{r}_{\text{DY}} \sim \mathbb{N}(0.1, 0.1)$ is entered and treated as independent of the HCare view under correlation agnosticism. MTE then generates the DY-mimicking portfolio, and SAR overlays it with the initial HCare tilt to form a composite factor tilt.

Step 4 *Analytic Feedback*. Factor exposures now show positive loadings on both DY and HCare, though DY remains smaller than HCare (Figure 8).

Step 5 *View-to-Portfolio Mapping*. To balance target exposures, the investor adjusts view uncertainties, with a lower uncertainty for the DY view representing stronger conviction. Portfolio massage enables investors to use these uncertainties as the degrees of freedom to align their *tactical* view portfolios with their *strategic* goals (Proposition 4.2).

Step 6 *Analytic Feedback*. Through iterative fine-tuning, the investor reaches $\widetilde{r}_{\text{DY}} \sim \mathbb{N}(0.1, 0.078)$, where portfolio exposure to factors HCare and DY are equalised (Figure 9).

The discovery process may continue iteratively with efficiency fully preserved (Propositions 4.1 and 4.3), while Proposition 4.2 guarantees the feasibility of fully aligning the portfolio with the investor's true intentions. Once the ideal exposures are achieved, view ambiguities are naturally resolved, and the portfolio decision completed: the composite factor tilt (upper pane, Figure 10) with satisfactory trading turnover constitutes the 'execution list', while the overlaid portfolio (lower pane) yields the 'enhanced indexation' portfolio – fulfilling the *tactical* intent without compromising the *strategic* goal.

Note that throughout the process, correlation-unawareness does not prevent investor from discovering an ideal portfolio, since all they care about is welfare (e.g. factor exposures, transaction costs due to trading turnover etc.), not view accuracy.

6. Implications

Recognising SAR-based allocation and MTE-guided mimicking as the core principles of active portfolio construction yields both theoretical and practical benefits.

Modularisation, Scalability, and Transparency. The framework demystifies portfolio construction by decomposing it into *allocation* and *mimicking* sub-problems solved via SAR and MTE. For instance, sophisticated factor management uses MTE for factor mimicking and SAR for allocations (Section 5). SAR also democratises investing: when stock differentiation is limited, it produces equal-weighted strategies, while weighting funds by manager skill yields fund-of-funds strategies.

Dimension Reduction, Comprehensiveness, and Unification. Predicting the performance of every security is no longer necessary. Investors can instead form views on broad market variables – *securities*, *factors*, *themes*, or *idiosyncratic returns* – which the framework then translates coherently into optimal portfolios according to the same allocation and mimicking principles.

Embracing Qualitative, Subjective, and Ambiguous Information. Portfolio massage allows correlated views to be treated as independent (Lemma 4.5), aligns ambiguous inputs with *strategic* goals (Proposition 4.2), and enables investors

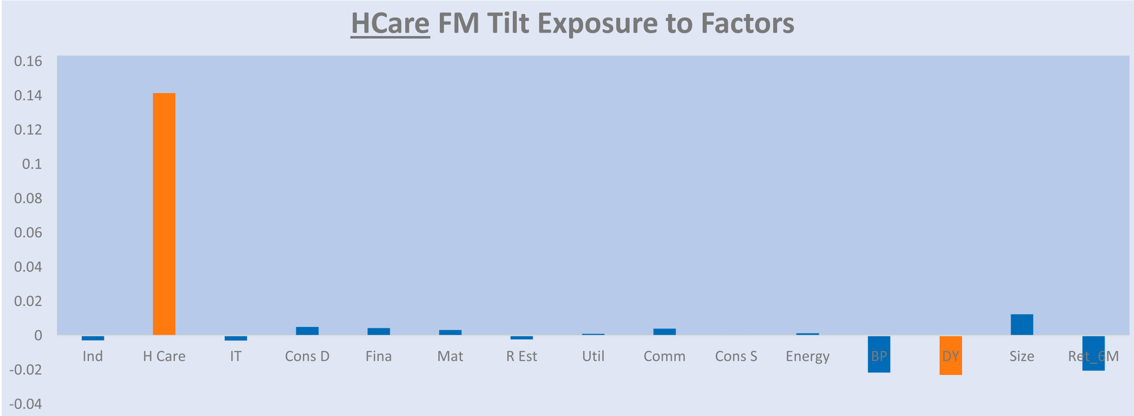

Figure 7. Analytical feedback: intended vs. unintended exposures.

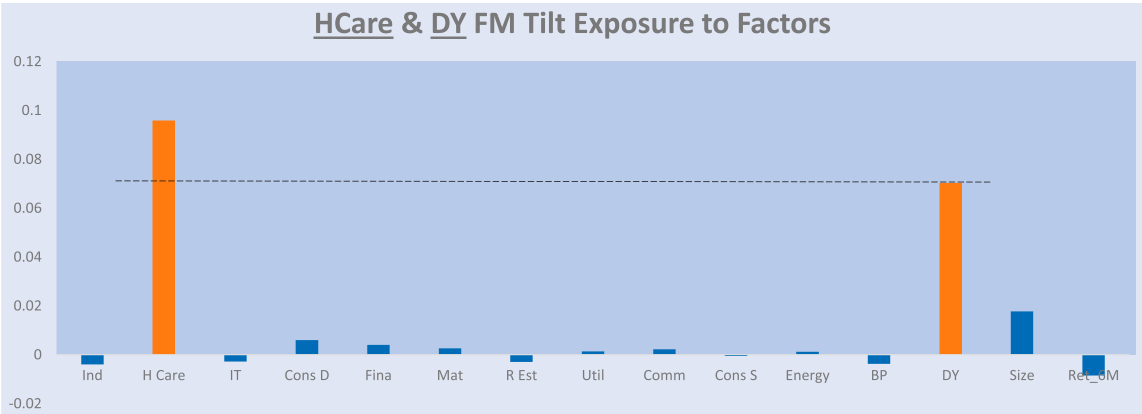

Figure 8. Analytical feedback: improved but still not ideal exposures.

of all sophistication levels to incorporate virtually any information – quantitative or qualitative, objective or subjective – into portfolio decisions.

From Complex Objective Settings to Intuitive View Expressions. Traditional optimisation requires explicitly pre-defining utility functions, which can be cumbersome. In contrast, the framework incorporates natural investor views, dynamically generates an implicit Bayesian utility, and outputs the optimal portfolio regardless of objective number or complexity.

Objective Implementation of Subjective Views. By admitting diverse views while unifying efficient view-to-portfolio mapping (Theorem 4.1), the framework divides labour: humans focus on the *art* of view formation, while the framework handles the *science* of portfolio generation, fulfilling Treynor and Black (1973)'s call to bridge *subjective* views with PS *objectivity*. Analytics-guided massage models portfolio selection as a *view-driven, feedback-responsive* goal discovery process.

Cognition-Friendly Portfolio Selection. Traditional one-shot optimisation is cognitively demanding. This framework converts investors' preliminary views into a seed portfolio and employs analytical feedback to refine it towards a balanced portfolio (Proposition 4.2; Section 5). Shifting from a 'rocket-science' to a 'portfolio-discovery' mindset aligns with human

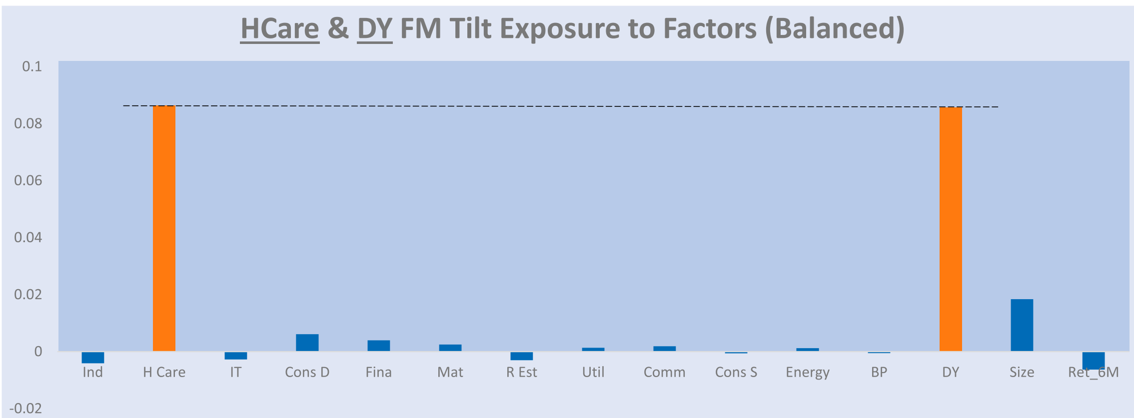

Figure 9. Analytical feedback: balanced tilt factor exposures.

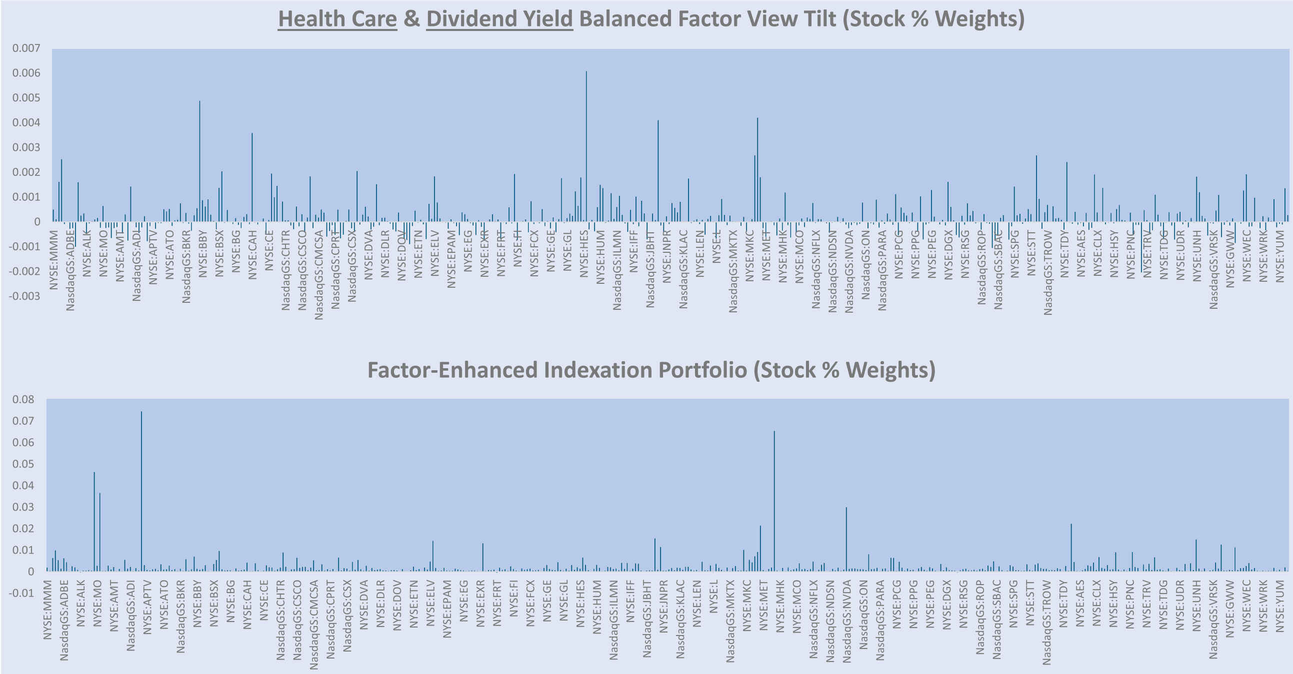

Figure 10. The final tilt & index-enhanced portfolio.

cognition and industry practice, with important theoretical and practical implications.

Reconciling Normative and Descriptive Portfolio Decision Making. The Bayesian Allocation Framework reflects *descriptive* decision-making by capturing real-world behaviours in one equation (17) and preserves *normative* character by internalising investor views into an evolving implicit Bayesian utility.

Machine Learning (ML)-Assisted Portfolio Discovery. While ML advances in *alpha modelling* (Heaton *et al.* 2016, Feng *et al.* 2019, Gu *et al.* 2020, Behera and Kumar 2025), *risk modelling* (Bien and Tibshirani 2011), and *trading policy learning* (Moody and Saffell 2001, Li *et al.* 2017, Zhang *et al.* 2025) enhance prediction, estimation, and automation, human-driven, machine-assisted methods remain vital in a regulated industry valuing transparency, diverse objectives, and personalisation. The framework promotes collaboration: ML generates signals while humans set goals, positioning ML as an enhancer of agile, scientifically robust, and philosophically aligned portfolios.

7. Conclusion

Active portfolio selection has long lacked a unified theory that reflects how real investors form and act on diverse beliefs. This research addresses that gap. This paper (I), the first of a two-part series, employs a mechanism-dissection methodology to show that two principles – the *Subjective Allocation Rule* (SAR) and the *Minimum Tracking Error* (MTE) – suffice to form a comprehensive *Bayesian Allocation Framework* unifying active portfolio-selection (PS) behaviours. Building on a mindset shift from objective mean-variance to subjective signal-noise efficiency, the two principles act synergistically: while MTE harnesses broad information sources, SAR embraces ambiguity and subjectivity to generate alpha (outperformance) proportional to investor forecasting *skill*. The framework integrates qualitative and quantitative beliefs about investor-chosen market variables, turning information ambiguities into flexibilities for aligning *tactical* views with *strategic* goals. By internalising investor beliefs into a dynamic utility, it reconciles *normative* rationality with *descriptive* realism: fragmented active PS behaviours naturally emerge as *local optima*, shaped by individual cognition, skill, and subjectivity. These insights reformulate active PS from a form of *rocket science* into a cognitively aligned *discovery process*, bridging theory and practice.

Practically, with the framework guaranteeing the *science* of portfolio implementation and feedback, investors can focus on the *art* of view formulation – or intent discovery. Accuracy in estimating the entire investment universe is no longer required; instead, investors have the freedom to form opinions on a flexible number of familiar subjects, whether tradable or not. Whether those inputs are quantitative or qualitative, independent or correlated, *analytics-guided portfolio massage* steers them towards portfolios that optimally balance investors' multifaceted considerations while maintaining sharpness in alpha generation. This makes active portfolio selection significantly simpler, more intuitive, transparent, and democratic across all skill levels.

The contributions of this paper are fourfold: (1) *Unifying Principles*: Identifies a minimal set of efficiency principles that unify investors' heterogeneous *active* PS behaviours under a single equation. (2) *Mechanism Extraction*: Reveals the underlying efficiency, alpha-generation, and information-processing mechanisms driving performance. (3) *Cognition Friendliness*: Establishes a scalable paradigm incorporating non-numeric, subjective, and ambiguous beliefs, as well as heterogeneous factor sensitivities, reflecting bounded rationality and dynamic utility. (4) *Methodologies*: Introduces a

mechanism-dissection approach to uncovering the inner workings of complex models, and a **descriptive characterisation approach** to modelling view ambiguity and subjectivity.

Two projects are planned to complete this research. First, the forthcoming Paper (II) validates SAR and MTE via portfolio analytics and backtests with regular rebalancing, and develops techniques (e.g. *skill-based view simulations*, *comparative subjectivity testing*) to assess their efficiency, alpha-generation, and information-processing mechanisms. Second, this paper identifies SAR and MTE as the minimal principles unifying a subset of active PS behaviours. To achieve theoretical closure, Cheung (2026) extends the argument from *parsimony* to *completeness*, demonstrating the framework's ability to coherently explain, integrate, and guide the full spectrum of heterogeneous real-world practices, including heuristics and folk wisdom.

Overall, the Bayesian Allocation Framework recasts active portfolio selection as an *intuitive discovery process*, unifying fragmented practices and transcending prescriptive, normative black boxes towards a form of *view-driven, feedback-responsive* rationality akin to backward induction as implemented in AI, albeit in a small-data setting. This, in turn, unlocks new frontiers for innovation in active portfolio management.

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Author contributions

CRediT: **Wing Cheung**: Writing – original draft, Writing – review & editing

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ORCID

Wing Cheung http://orcid.org/0000-0003-4730-4020

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Appendix. Proofs

A.1. Breaking down the ABL model model

The ABL model takes the following form:

THEOREM A.1 (The ABL Model) *The posterior return estimates are normally distributed, that is,* $\widetilde{\boldsymbol{r}}^a_{|\widehat{\boldsymbol{q}}^a,\mathcal{H},\mathcal{G}} \sim \mathbb{N}(\widehat{\boldsymbol{m}}^a, \widehat{\mathbf{V}}^a)$, *where the* $[N \times 1]$ *updated mean vector is:*

$$\widehat{\boldsymbol{m}}^a = \left[(\tau\boldsymbol{\Sigma}^a)^{-1} + (\mathbf{P}^a)^{\mathrm{T}}(\boldsymbol{\Omega}^a)^{-1}\mathbf{P}^a\right]^{-1} \times \left[(\tau\boldsymbol{\Sigma}^a)^{-1}\widehat{\boldsymbol{\pi}}^a + (\mathbf{P}^a)^{\mathrm{T}}(\boldsymbol{\Omega}^a)^{-1}\widehat{\boldsymbol{q}}^a\right] \tag{A1}$$

and the $[N \times N]$ *updated variance-covariance matrix is:*

$$\widehat{\mathbf{V}}^a = \left[(\tau\boldsymbol{\Sigma}^a)^{-1} + (\mathbf{P}^a)^{\mathrm{T}}(\boldsymbol{\Omega}^a)^{-1}\mathbf{P}^a\right]^{-1} \tag{A2}$$

Proof See Appendix A to Cheung (2013). ■

This model can be decomposed as follows:

LEMMA A.1 (The ABL Model Decomposition) *The posterior return estimates are Normally distributed, i.e.* $\widetilde{\boldsymbol{r}}^a_{|\widehat{\boldsymbol{q}}^a,\mathcal{H},\mathcal{G}} \sim \mathbb{N}(\widehat{\boldsymbol{m}}^a_{[N\times 1]}, \widehat{\mathbf{V}}^a_{[N\times N]})$, *where the updated mean estimates are:*

$$\widehat{\boldsymbol{m}}^a = \widehat{\mathbf{V}}^a \left\{ \frac{\kappa}{\tau}\begin{pmatrix} \boldsymbol{w}_M \\ \mathbf{0}_{[f\times 1]} \end{pmatrix} + \left[\begin{pmatrix} \mathbf{I}_n \\ \mathbf{0}_{[f\times n]} \end{pmatrix} \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} + \begin{pmatrix} \mathbf{0}_{[n\times f]} \\ \mathbf{I}_{[f\times f]} \end{pmatrix} \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\widehat{\boldsymbol{q}}_F + \begin{pmatrix} \mathbf{I}_n \\ -\mathbf{B}^{\mathrm{T}}_{[f\times n]} \end{pmatrix} \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi \right]\right\} \tag{A3}$$

and the updated variance-covariance matrix is:

$$\widehat{\mathbf{V}}^a = \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{12}^{\mathrm{T}} & \mathbf{M}_{22} \end{pmatrix} \tag{A4}$$

where

$$\mathbf{M}_{11[n\times n]} = \left[\left(\boldsymbol{\Sigma}^+\right)^{-1} + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1} \tag{A5}$$

$$\mathbf{M}_{12[n\times f]} = \left[\left(\boldsymbol{\Sigma}^+\right)^{-1} + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1}\left(\boldsymbol{\Sigma}^+\right)^{-1}\mathbf{B}\boldsymbol{\Sigma}_F^+ = \mathbf{M}_{11}\left(\boldsymbol{\Sigma}^+\right)^{-1}\mathbf{B}\boldsymbol{\Sigma}_F^+ \tag{A6}$$

$$\mathbf{M}_{22[f\times f]} = \left[\left(\boldsymbol{\Sigma}_F^+\right)^{-1} + (\mathbf{PB})^{\mathrm{T}}\left(\mathbf{P}\boldsymbol{\Sigma}_\xi^+\mathbf{P}^{\mathrm{T}} + \boldsymbol{\Omega}\right)^{-1}(\mathbf{PB})\right]^{-1} \tag{A7}$$

$$\boldsymbol{\Sigma}^+_{[n\times n]} = \mathbf{B}\boldsymbol{\Sigma}_F^+\mathbf{B}^{\mathrm{T}} + \boldsymbol{\Sigma}_\xi^+ \tag{A8}$$

$$\boldsymbol{\Sigma}^+_{F[f\times f]} = \left[(\tau\boldsymbol{\Sigma}_F)^{-1} + \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\mathbf{P}_F\right]^{-1} \tag{A9}$$

$$\boldsymbol{\Sigma}^+_{\xi[n\times n]} = \left[(\tau\boldsymbol{\Sigma}_\xi)^{-1} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi\right]^{-1} \tag{A10}$$

Proof It is easy to validate:

$$(\boldsymbol{\Sigma}^a)^{-1} = \begin{bmatrix} (\boldsymbol{\Sigma}_\xi)^{-1} & -(\boldsymbol{\Sigma}_\xi)^{-1}\mathbf{B} \\ -\mathbf{B}^{\mathrm{T}}(\boldsymbol{\Sigma}_\xi)^{-1} & (\boldsymbol{\Sigma}_F)^{-1} + \mathbf{B}^{\mathrm{T}}(\boldsymbol{\Sigma}_\xi)^{-1}\mathbf{B} \end{bmatrix}; \tag{A11}$$

$$(\mathbf{P}^a)^{\mathrm{T}}(\boldsymbol{\Omega}^a)^{-1}\mathbf{P}^a = \begin{bmatrix} \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi & -\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi\mathbf{B} \\ -\mathbf{B}^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi & \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\mathbf{P}_F + \mathbf{B}^{\mathrm{T}}(\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi)\mathbf{B} \end{bmatrix};$$

and (A12)

$$(\mathbf{P}^a)^{\mathrm{T}}(\boldsymbol{\Omega}^a)^{-1}\widehat{\boldsymbol{q}}^a = \begin{pmatrix} \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi \\ \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\widehat{\boldsymbol{q}}_F - \mathbf{B}^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi \end{pmatrix}. \tag{A13}$$

$$\begin{aligned}(\tau\boldsymbol{\Sigma}^a)^{-1}\widehat{\boldsymbol{\pi}}^a &\overset{\because \widehat{\boldsymbol{\pi}}=\kappa\boldsymbol{\Sigma}\boldsymbol{w}_M}{\underset{\widehat{\boldsymbol{\pi}}_F=\kappa\boldsymbol{\Sigma}_F\mathbf{B}^{\mathrm{T}}\boldsymbol{w}_M}{=}} \frac{\kappa}{\tau}\begin{bmatrix} (\boldsymbol{\Sigma}_\xi)^{-1} & -(\boldsymbol{\Sigma}_\xi)^{-1}\mathbf{B} \\ -\mathbf{B}^{\mathrm{T}}(\boldsymbol{\Sigma}_\xi)^{-1} & (\boldsymbol{\Sigma}_F)^{-1} + \mathbf{B}^{\mathrm{T}}(\boldsymbol{\Sigma}_\xi)^{-1}\mathbf{B} \end{bmatrix} \times \begin{pmatrix} \boldsymbol{\Sigma} \\ \boldsymbol{\Sigma}_F\mathbf{B}^{\mathrm{T}} \end{pmatrix}\boldsymbol{w}_M \\ &= \frac{\kappa}{\tau}\begin{pmatrix} \boldsymbol{w}_M \\ \mathbf{0}_{[f\times 1]} \end{pmatrix}.\end{aligned} \tag{A14}$$

Plugging (A14) and (A13) into (A1), and noting (A2) yields:

$$\widehat{\boldsymbol{m}}^a = \widehat{\mathbf{V}}^a \begin{pmatrix} \frac{\kappa}{\tau}\boldsymbol{w}_M + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi \\ \mathbf{0}_{[f\times 1]} + \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\widehat{\boldsymbol{q}}_F - \mathbf{B}^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi \end{pmatrix}. \tag{A15}$$

This is equivalent to (A3).

Substituting (A11) and (A12) into (A2), we obtain (A4):

$$\begin{aligned}\widehat{\mathbf{V}}^a &= \left\{\frac{1}{\tau}\begin{bmatrix} (\boldsymbol{\Sigma}_\xi)^{-1} & -(\boldsymbol{\Sigma}_\xi)^{-1}\mathbf{B} \\ -\mathbf{B}^{\mathrm{T}}(\boldsymbol{\Sigma}_\xi)^{-1} & (\boldsymbol{\Sigma}_F)^{-1} + \mathbf{B}^{\mathrm{T}}(\boldsymbol{\Sigma}_\xi)^{-1}\mathbf{B} \end{bmatrix} + \begin{bmatrix} \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi & -\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi\mathbf{B} \\ -\mathbf{B}^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi & \mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\mathbf{P}_F + \mathbf{B}^{\mathrm{T}}(\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\mathbf{P}_\xi)\mathbf{B} \end{bmatrix}\right\}^{-1} \\ &= \begin{pmatrix} \left[\left(\boldsymbol{\Sigma}^+\right)^{-1} + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1} & \left[\left(\boldsymbol{\Sigma}^+\right)^{-1} + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1}\left(\boldsymbol{\Sigma}^+\right)^{-1}\mathbf{B}\boldsymbol{\Sigma}_F^+ \\ \boldsymbol{\Sigma}_F^+\mathbf{B}^{\mathrm{T}}\left(\boldsymbol{\Sigma}^+\right)^{-1}\left[\left(\boldsymbol{\Sigma}^+\right)^{-1} + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1} & \left[\left(\boldsymbol{\Sigma}_F^+\right)^{-1} + (\mathbf{PB})^{\mathrm{T}}\left(\mathbf{P}\boldsymbol{\Sigma}_\xi^+\mathbf{P}^{\mathrm{T}} + \boldsymbol{\Omega}\right)^{-1}(\mathbf{PB})\right]^{-1} \end{pmatrix}.\end{aligned} \tag{A16}$$

This concludes the proof. ■

A.2. Derivation of the transparent ABL allocation

The Transparent ABL Allocation is proved as follows:

THEOREM 3.1 (The Transparent ABL (TABL) Allocation) *In the absence of constraints, the mean-variance efficient ABL posterior allocation admits the following representation:*

$$\boldsymbol{w}^* = \underbrace{\underbrace{\frac{\kappa}{2\lambda\tau}\boldsymbol{w}_M}_{\text{benchmark}} + \underbrace{\mathbf{P}^{\mathrm{T}}(2\lambda\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}}}_{\text{security/strategy combinationtilt}}}_{\text{Original BL allocation}} + \underbrace{\underbrace{\mathbf{M}_{\mathrm{FM}}^{\mathrm{T}}\mathbf{P}_F^{\mathrm{T}}(2\lambda\boldsymbol{\Omega}_F)^{-1}\widehat{\boldsymbol{q}}_F}_{\text{composite factor}-\text{mimickingtilt}} + \underbrace{(\mathbf{I} - \mathbf{B}\mathbf{M}_{\mathrm{FM}})^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}\left(2\lambda\boldsymbol{\Omega}_\xi\right)^{-1}\widehat{\boldsymbol{q}}_\xi}_{\text{idiosyncratic tilt}}}_{\text{Augmented tilts inABL}}. \tag{A17}$$

Proof Lemma A.1 gives the posterior joint distribution of the n securities and f factors in the augmented universe. For whatever reasons, if factors are uninvestible, we need to construct a pure security portfolio based on the n-dimensional posterior *marginal distribution* of the securities. This can be legitimately represented by the simple 'carve-outs':

$$\widehat{\boldsymbol{m}}_{[n\times 1]} = \mathbf{M}_{11}\left[\left(\frac{\kappa}{\tau}\boldsymbol{w}_M + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} + \mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi\right) + \left[(\boldsymbol{\Sigma}^+)^{-1}\mathbf{B}\boldsymbol{\Sigma}_F^+\right]\left(\mathbf{P}_F^{\mathrm{T}}\boldsymbol{\Omega}_F^{-1}\widehat{\boldsymbol{q}}_F - \mathbf{B}^{\mathrm{T}}\mathbf{P}_\xi^{\mathrm{T}}\boldsymbol{\Omega}_\xi^{-1}\widehat{\boldsymbol{q}}_\xi\right)\right]; \tag{A18}$$

and

$$\widehat{\mathbf{V}}_{[n\times n]} = \mathbf{M}_{11} = \left[\left(\boldsymbol{\Sigma}^+\right)^{-1} + \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{P}\right]^{-1}. \tag{A19}$$

In the absence of optimisation constraints, the posterior mean-variance optimisation problem can generally be written as:

$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{\operatorname{argmax}} \left\{ \boldsymbol{w}^{\mathrm{T}} \widehat{\boldsymbol{m}} - \lambda \boldsymbol{w}^{\mathrm{T}} \widehat{\mathbf{V}} \boldsymbol{w}_{[n\times 1]} \right\}. \quad \text{(A20)}$$

This simple setting permits the following analytical allocation solution:

$$\begin{aligned} \boldsymbol{w}^* &= \frac{1}{2\lambda} \widehat{\mathbf{V}}^{-1} \widehat{\boldsymbol{m}} \\ &= \frac{\kappa}{2\lambda\tau} \boldsymbol{w}_M + \mathbf{P}^{\mathrm{T}} (2\lambda \boldsymbol{\Omega})^{-1} \widehat{\boldsymbol{q}} \\ &\quad + \left[(\boldsymbol{\Sigma}^+)^{-1} \mathbf{B} \boldsymbol{\Sigma}_F^+ \right] \mathbf{P}_F^{\mathrm{T}} (2\lambda \boldsymbol{\Omega}_F)^{-1} \widehat{\boldsymbol{q}}_F \\ &\quad + \left[(\boldsymbol{\Sigma}^+)^{-1} \boldsymbol{\Sigma}_\xi^+ \right] \mathbf{P}_\xi^{\mathrm{T}} (2\lambda \boldsymbol{\Omega}_\xi)^{-1} \widehat{\boldsymbol{q}}_\xi. \end{aligned} \quad \text{(A21)}$$

With notation (18), this is equivalent to (A17). ■

A.3. Minimum-Tracking-Error (MTE) principle

PROPOSITION 4.1 (Principle 1: MTE-Guided Mimicking) *ABL internally pursues the Minimum-Tracking-Error (MTE) principle consistently to guarantee efficient FM, SSM and hedging.*

Proof Since the factor model (1) holds in the 'posterior' market as well, we have:

$$\boldsymbol{\Sigma}^+ = \mathbf{B} \boldsymbol{\Sigma}_F^+ \mathbf{B}^{\mathrm{T}} + \boldsymbol{\Sigma}_\xi^+. \quad \text{(A22)}$$

Given a factor portfolio $\boldsymbol{w}_F$, we form the following minimum-variance tracking problem (equivalent to the MTE problem):

$$\left(\boldsymbol{w}^{[\mathrm{T}]} \right)^* = \underset{\boldsymbol{w}^{[\mathrm{T}]}}{\operatorname{argmin}} \, \mathbb{V}ar \left[\left(\boldsymbol{w}^{[\mathrm{T}]} \right)^{\mathrm{T}} \tilde{\boldsymbol{r}} - \boldsymbol{w}_F^{\mathrm{T}} \tilde{\boldsymbol{r}}_F \right]. \quad \text{(A23)}$$

where $\boldsymbol{w}^{[\mathrm{T}]}$ is the $[n \times 1]$ security-level weight vector of the tracking portfolio; and $\mathbb{V}ar(\cdot)$ denotes the variance operator. This problem aims to find the security-level portfolio $\boldsymbol{w}^{[\mathrm{T}]}$ whose performance tracks that of the factor portfolio $\boldsymbol{w}_F$ with minimum tracking errors.

The solution to this problem happens to be:

$$\left(\boldsymbol{w}^{[\mathrm{T}]} \right)^* = \left[(\boldsymbol{\Sigma}^+)^{-1} \mathbf{B} \boldsymbol{\Sigma}_F^+ \right] \boldsymbol{w}_F = \mathbf{M}_{\mathrm{FM}}^{\mathrm{T}} \boldsymbol{w}_F = \boldsymbol{w}^{[\mathrm{FM}]}. \quad \text{(A24)}$$

So under the MTE principle, $\mathbf{M}_{\mathrm{FM}}^{\mathrm{T}}$ is indeed the *FM transformation matrix* or *technique*. Given a factor strategy $\boldsymbol{w}_F$, its security-level FM portfolio $\mathbf{M}_{\mathrm{FM}}^{\mathrm{T}} \boldsymbol{w}_F$ is expected to track the performance of the factor strategy subject to minimum tracking errors:

$$\mathbb{E} \left[\boldsymbol{w}_F^{\mathrm{T}} \mathbf{M}_{\mathrm{FM}} \tilde{\boldsymbol{r}} \right] \overset{\mathrm{MTE}}{=} \mathbb{E} \left[\boldsymbol{w}_F^{\mathrm{T}} \tilde{\boldsymbol{r}}_F \right], \quad \text{(A25)}$$

where '$\overset{\mathrm{MTE}}{=}$' denotes equality subject to minimum tracking errors in a stochastic performance-mimicking sense.

It can be demonstrated in the same way that $(\mathbf{I}_n - \mathbf{B}\mathbf{M}_{\mathrm{FM}})^{\mathrm{T}} = [(\boldsymbol{\Sigma}^+)^{-1} \boldsymbol{\Sigma}_\xi^+]$ is the *SSM transformation matrix* or *technique* under the MTE principle. Given a security idiosyncratic strategy $\boldsymbol{w}_\xi$, its security-level SSM portfolio $(\mathbf{I}_n - \mathbf{B}\mathbf{M}_{\mathrm{FM}})^{\mathrm{T}} \boldsymbol{w}_\xi$ is expected to track the performance of the security idiosyncratic strategy subject to minimum tracking errors:

$$\mathbb{E} \left[\boldsymbol{w}_\xi^{\mathrm{T}} (\mathbf{I}_n - \mathbf{B}\mathbf{M}_{\mathrm{FM}}) \tilde{\boldsymbol{r}} \right] \overset{\mathrm{MTE}}{=} \mathbb{E} \left[\boldsymbol{w}_\xi^{\mathrm{T}} \tilde{\boldsymbol{\xi}} \right]. \quad \text{(A26)}$$

Lastly, given a security portfolio $\boldsymbol{w}$, if we want to hedge its common risks, we seek a tracking portfolio for the systematic return of $\boldsymbol{w}$, that is, $\boldsymbol{w}^{\mathrm{T}} \mathbf{B} \tilde{\boldsymbol{r}}_F$. Since $(\mathbf{B}\mathbf{M}_{\mathrm{FM}})^{\mathrm{T}} = (\boldsymbol{\Sigma}^+)^{-1} \mathbf{B} \boldsymbol{\Sigma}_F^+ \mathbf{B}^{\mathrm{T}}$ is the transformation matrix solution to the corresponding MTE problem, it is the *hedging transformation matrix* or *technique* under the MTE principle. ■

A.4. SAR alpha-generating mechanism

LEMMA 4.2 (SAR Alpha Generation with Orthogonal Views) *SAR generates an information ratio (IR) directly proportional to view skill (IC) and the betting scale. When views are both skilled and mutually orthogonal, SAR guarantees aggregate alpha, without compounding errors.*

Proof Since SAR enforces consistent allocation across all levels, we may, without loss of generality, focus on security- or portfolio-level views; as noted in Figure 2, 2λ is omitted. Using the diagonalisation in (23), we examine the expected performance of the SAR tilt:

$$\mathbb{E} \left(\tilde{\boldsymbol{y}}^{\mathrm{T}} \boldsymbol{\Omega}^{-1} \tilde{\boldsymbol{x}} \right) = \mathbb{E} \left[(\tilde{\boldsymbol{y}}^\perp)^{\mathrm{T}} (\boldsymbol{\Omega}^\perp)^{-1} \tilde{\boldsymbol{x}}^\perp \right] = \sum_{i=1}^{k_1} \frac{\mathbb{E} \left(\tilde{y}_i^\perp \tilde{x}_i^\perp \right)}{(\omega_i^\perp)^2} \quad \text{(A27)}$$

where $\tilde{x}_i^\perp$ is the i^{th} return element of $\tilde{\boldsymbol{x}}^\perp = \mathbf{P}^\perp \tilde{\boldsymbol{r}}$; and $\tilde{y}_i^\perp$ is the ith view element of $\tilde{\boldsymbol{y}}^\perp$.

At portfolio selection, (A27) is evaluated using public information $\mathcal{G}$ and private information $\mathcal{H}$. Public information $\mathcal{G}$ provides analysts' track records for assessing forecasting skill, while private information $\mathcal{H}$ yields analyst view updates $\tilde{\boldsymbol{y}}_{|\mathcal{H},\mathcal{G}}$. To be clear, since $\mathcal{H}$ is private, it enters only *ex ante* via $\tilde{\boldsymbol{y}}_{|\mathcal{H},\mathcal{G}}$, not *ex post* $\tilde{\boldsymbol{x}}_{|\mathcal{G}}$. Accordingly, posterior updates of $\tilde{\boldsymbol{x}}$ under $\mathcal{H}$ are inferred *indirectly* from historical forecasting skill and current view updates.

Given $\mathcal{G}$, assume the ith view $\tilde{y}_i^\perp$ and the corresponding strategy's active return $\tilde{x}_i^\perp$ are jointly distributed as follows:

$$\begin{pmatrix} \tilde{y}_i^\perp \\ \tilde{x}_i^\perp \end{pmatrix} \overset{|\mathcal{G}}{\sim} \mathbb{N} \left[\begin{pmatrix} \mu_{y_i} \\ \mu_{x_i} \end{pmatrix}, \begin{pmatrix} \sigma_{y_i}^2 & \rho_i \sigma_{y_i} \sigma_{x_i} \\ \rho_i \sigma_{y_i} \sigma_{x_i} & \sigma_{x_i}^2 \end{pmatrix} \right] \quad \text{(A28)}$$

where $\mu_{y_i} = \mathbb{E}(\tilde{y}_i^\perp \,|\, \mathcal{G})$; $\mu_{x_i} = \mathbb{E}(\tilde{x}_i^\perp \,|\, \mathcal{G})$; $\sigma_{y_i}^2 = \mathbb{V}ar(\tilde{y}_i^\perp \,|\, \mathcal{G})$; $\sigma_{x_i}^2 = \mathbb{V}ar(\tilde{x}_i^\perp \,|\, \mathcal{G})$; $\rho_i = \mathbb{C}orr(\tilde{y}_i^\perp, \tilde{x}_i^\perp \,|\, \mathcal{G})$.

The joint distribution captures the historical relationship between the *ex ante* view $(\tilde{y}_i^\perp)$ and the *ex post* realised active return $\tilde{x}_i^\perp$ of strategy i, based on analysts' track records in $\mathcal{G}$. The correlation, ρ_i, defines the analysts' *forecasting skill*, or *information coefficient* (IC) (Jensen, 1972, cited in Henriksson and Merton (1981)). As views imply *active* investing, performance is measured in *active* space relative to the market. Under *semi-strong market efficiency*, $\mu_{y_i} = \mu_{x_i} = 0$.

Upon arrival of $\mathcal{H}$, analysts update their views to point estimates, $\tilde{\boldsymbol{y}}_{|\mathcal{H},\mathcal{G}} \overset{\text{belief}}{\to} \widehat{\boldsymbol{q}}$, leading to the following aggregate view outperformance evaluation of the SAR tilt:

$$\begin{aligned} \alpha &= \mathbb{E} \left(\tilde{\boldsymbol{y}}^{\mathrm{T}} \boldsymbol{\Omega}^{-1} \tilde{\boldsymbol{x}} \,|\, \{\tilde{\boldsymbol{y}}_{\,|\,\mathcal{H},\mathcal{G}} = \widehat{\boldsymbol{q}}\}, \mathcal{H}, \mathcal{G} \right) \\ &= \sum_{i=1}^{k_1} \frac{\mathbb{E} \left(\tilde{y}_i^\perp \tilde{x}_i^\perp \,|\, \{\tilde{y}_{i|\mathcal{H},\mathcal{G}}^\perp = \widehat{q}_i^\perp\}, \mathcal{H}, \mathcal{G} \right)}{(\omega_i^\perp)^2} \\ &= \sum_{i=1}^{k_1} \frac{\mathbb{E} \left(\tilde{y}_{i|\mathcal{H},\mathcal{G}}^\perp \tilde{x}_{i|\mathcal{G}}^\perp \,|\, \{\tilde{y}_{i|\mathcal{H},\mathcal{G}}^\perp = \widehat{q}_i^\perp\} \right)}{(\omega_i^\perp)^2} \\ &= \sum_{i=1}^{k_1} \frac{\widehat{q}_i^\perp}{(\omega_i^\perp)^2} \left(\mu_{x_i} + \rho_i \frac{\sigma_{x_i}}{\sigma_{y_i}} (\widehat{q}_i^\perp - \mu_{y_i}) \right) \\ &= \sum_{i=1}^{k_1} \rho_i \frac{\sigma_{x_i}}{\sigma_{y_i}} \left(\frac{\widehat{q}_i^\perp}{\omega_i^\perp} \right)^2. \end{aligned} \quad \text{(A29)}$$

This implies that, for any individual view, positive skill corresponds to outperformance, zero skill to no added value, and negative skill to underperformance. By rearranging, it follows that SAR generates an information ratio (IR) directly proportional to skill (IC) and to the betting scale:

$$\underbrace{\frac{\alpha_i}{\sigma_{x_i}}}_{\text{IR}} = \underbrace{\rho_i}_{\text{skill IC}} \times \underbrace{\underbrace{\frac{\widehat{q}_i^{\perp}}{\sigma_{y_i}}}_{\text{signal strength}} \times \underbrace{\frac{\widehat{q}_i^{\perp}}{(\omega_i^{\perp})^2}}_{\text{view conviction}}}_{\text{betting scale}(>0)} . \tag{A30}$$

Because view correlations are unobservable, submitting views with high orthogonalised skill is cognitively infeasible. We therefore restrict attention to orthogonal views, anticipating that Subsection 4.4.3 will show that correlated views can always be equivalently represented in this form.

As shown in (A29), *under orthogonal views*, errors arising from individual views, mimicking, or hedging are *additive* – and therefore potentially diversifiable via the Law of Large Numbers – rather than multiplicative. Consequently, *SAR delivers aggregate risk-adjusted alpha without compounding errors*. ■

A.5. *Information-processing mechanisms*

LEMMA 4.3 (SAR Maps Indistinguishable Views to Distinct Portfolios) *SAR maps indistinguishable orthogonal directional views of varying strengths into analytically distinct portfolios.*

Proof Suppose an investor has k directional views processed as orthogonal (Assumption 4.2), $\widetilde{\boldsymbol{r}}_{[k\times1]} \sim \mathbb{N}(\widehat{\boldsymbol{q}}_{[k\times1]}, \boldsymbol{\Lambda}_{[k\times k]})$. Focusing on the jth ($1 \leqslant j \leqslant k$) view, $\widetilde{r}_j \sim \mathbb{N}(\widehat{q}_j, \omega_j^2)$, Assumption 4.1 implies the investor cannot distinguish it from any members of the family $\{\widetilde{r}_j \sim \mathbb{N}(\widehat{q}_j, \eta_j\omega_j^2) \mid \forall \eta_j > 0\}$. Thus, they perceive $\{\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\eta}\boldsymbol{\Lambda}) \mid \forall \boldsymbol{\eta} > 0\}$, where $\boldsymbol{\eta}$ is a $[k \times k]$ diagonal matrix with positive elements η_j, as a k-dimension family of *indistinguishable views*.

When $\boldsymbol{\eta} \not\propto \mathbf{I}_k$ (*inhomogeneous* view scaling), SAR maps this family to an infinite set of linearly independent portfolios up to scaling: $\boldsymbol{w}^* \propto (\boldsymbol{\eta}\boldsymbol{\Lambda})^{-1}\widehat{\boldsymbol{q}} \not\propto \boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{q}}$. Each portfolio corresponds to different relative view strengths.

While views are *cognitive* and *ambiguous*, the resulting portfolios are *objective* and *tangible*. Differences between linearly independent portfolios can be detected through portfolio analytics. Therefore, SAR can transform cognitively or subjectively *indistinguishable directional views* with varying relative strengths into *analytically distinct portfolios*. ■

LEMMA 4.4 (Representation of Orthogonal Directional Views) *Portfolio massage sufficiently operationalises orthogonal directional views.*

Proof Since portfolio massage allows direct subjective adjustments of the diagonal elements of the view uncertainty matrix $\boldsymbol{\Lambda}$, any directional view family $\{\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\eta}\boldsymbol{\Lambda}) \mid \forall \boldsymbol{\eta} > 0\}$ spanned through $\boldsymbol{\eta}$-scaling can also be attained by $\boldsymbol{\Lambda}$-massage. Therefore, portfolio massage sufficiently represents *orthogonal directional* views (Assumptions 4.1 and 4.2). ■

LEMMA 4.5 (Orthogonal Representation of Correlated Views) *Under SAR, correlated views can always be represented welfare-equivalently by orthogonal views through portfolio massage.*

Proof For simplicity, consider k_1 security-level views $\mathbf{P}\widetilde{\boldsymbol{r}} \sim \mathbb{N}(\widehat{\boldsymbol{q}}, \boldsymbol{\Omega})$, where $\boldsymbol{\Omega}$ is the real, unknown view uncertainty matrix with non-zero off-diagonal entries. Assuming humans are unaware of correlations and use a $[k_1 \times k_1]$ diagonal matrix $\boldsymbol{\Lambda}$ to represent $\boldsymbol{\Omega}$, it suffices to show that $\mathbf{P}^{\mathrm{T}}(\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}}$ can be reachable via $\mathbf{P}^{\mathrm{T}}(\boldsymbol{\Lambda})^{-1}\widehat{\boldsymbol{q}}$ by portfolio massage, that is, adjusting the diagonal entries of $\boldsymbol{\Lambda}$, or the view uncertainty parameters.

Notate:

$$\boldsymbol{\Omega}^{-1}\widehat{\boldsymbol{q}} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_{k_1} \end{pmatrix}, \tag{A31}$$

where b_i is the ith view tilt (weight). As long as we, through portfolio massage, reach:

$$\boldsymbol{\Lambda} = \begin{pmatrix} \dfrac{\widehat{q}_1}{b_1} & 0 & \cdots & 0 \\ 0 & \dfrac{\widehat{q}_2}{b_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \dfrac{\widehat{q}_{k_1}}{b_{k_1}} \end{pmatrix}, \tag{A32}$$

we have $(\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}} = (\boldsymbol{\Lambda})^{-1}\widehat{\boldsymbol{q}}$.

This means any correlated set of views admits an equivalent orthogonal representation, which SAR maps to the same portfolios. Since views are *subjective* and *ambiguous*, but portfolios determine investor *welfare*, investors are indifferent between the original and orthogonal view sets.

The only caveat is that the above proof does not guarantee positive-definiteness of the view uncertainty matrix $\boldsymbol{\Lambda}$. However, this does not undermine practical implementation, as the incremental screening approach introduced below systematically incorporates relatively independent, mutually value-adding views without compromising alpha:

(1) *Pre-screening*: An experienced PM can prevent excessively correlated views from entering the same portfolio.
(2) *Filtering*: Pre-screened views are added one by one. If two highly correlated views i and j enter, abnormal portfolio behaviour (e.g. excessive factor exposure) can be detected via portfolio analytics. Portfolio massage then weakens one view, say j. Since negative uncertainty is prohibited, ω_j^2 is pushed to ∞, effectively removing j, while i remains in subsequent rounds. This process continues until a relatively orthogonal view set is obtained.

Consequently, correlated views can always be welfare-equivalently represented by orthogonal views. ■

PROPOSITION 4.2 (Feasibility of Pareto Exposure Improvements) *Under SAR, portfolio massage leverages additional information to quantify and implement ambiguous views, converting ambiguity into Pareto improvements aligned with strategic goals while preserving tactical views. Improvements can be manual, but well-defined exposure objectives enable optimisation.*

Proof Because the investor is correlation-agnostic (Assumption 4.2), the k_1 security- or portfolio-level views (as in (20)) are generally treated as orthogonal. SAR yields:

$$\boldsymbol{w} \propto \mathbf{P}^{\mathrm{T}}(\boldsymbol{\Omega})^{-1}\widehat{\boldsymbol{q}} = \frac{\widehat{q}_1}{\omega_1^2}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^2}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2}\boldsymbol{p}_{k_1} \tag{A33}$$

By Assumption 4.1, $\boldsymbol{\Omega}$'s diagonal elements ω_j^2 ($1 \leqslant j \leqslant k_1$) represent the investor's initial ambiguous view uncertainties, adjustable via portfolio massage. Treating these k_1 degrees of freedom as active levers, investors can actively calibrate view uncertainties to generate analytically distinct portfolios (Lemma 4.3), with portfolio analytics providing feedback to guide improved exposures.

To illustrate, portfolio exposures at the security, factor, theme or security-specific level can be written as:

$$\boldsymbol{e} = \boldsymbol{f}\left(\frac{\widehat{q}_1}{\omega_1^2}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^2}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2}\boldsymbol{p}_{k_1}\right) \tag{A34}$$

where $\boldsymbol{e} = (e_1, e_2, \ldots, e_m)^{\mathrm{T}}$ is the $[m \times 1]$ vector of analytical exposure measures (e.g. portfolio beta, factor exposure, risk decomposition, or conditional value-at-risk (CVAR) etc.); m stands for the number of exposure measures that the investor cares about; $\boldsymbol{f} = (f_1, f_2, \ldots, f_m)^{\mathrm{T}}$ are m portfolio-to-exposure transformation functions, each corresponding to one exposure measure.

Given an initial portfolio $\boldsymbol{w}$ (A33), the portfolio-to-exposure mapping (A34) yields initial analytical feedback $\boldsymbol{e}$, which can be visualised. Suppose the investor's ideal exposure is $\boldsymbol{e}^*$. If new uncertainties $\omega_j'^2$ $(1 \leqslant j \leqslant k_1)$ are identified such that the resulting exposure $\boldsymbol{e}'$ approaches $\boldsymbol{e}^*$, *Pareto improvements* are achieved, since these adjustments enhance the exposure profile without compromising view implementation (the investor is indifferent between ω_j^2 and $\omega_j'^2$).

This portfolio massaging problem can be formulated as an optimisation problem:

$$\left((\omega_1^2, \omega_2^2, \ldots, \omega_{k_1}^2)^{\mathrm{T}}\right)^* = \underset{(\omega_1^2, \omega_2^2, \ldots, \omega_{k_1}^2)^{\mathrm{T}}}{\operatorname{argmin}} \left\| \boldsymbol{e}^* - \boldsymbol{f}\left(\frac{\widehat{q}_1}{\omega_1^2}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^2}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2}\boldsymbol{p}_{k_1}\right)\right\|_2^2, \tag{A35}$$

which selects view uncertainty parameters ω_j^{*2} $(1 \leqslant j \leqslant k_1)$ to minimise the distance between the portfolio exposure profile $\boldsymbol{e}$ and the ideal profile $\boldsymbol{e}^*$. Given that $\boldsymbol{f}$ are typically well-defined and smooth, and that linearly dependent view strategies $\boldsymbol{p}_j$ can be avoided, this problem is numerically tractable. Hence, Pareto exposure improvements relative to $\boldsymbol{e}^*$ exist through portfolio massage.

In case $\boldsymbol{f}$ can be linearised, exposures can be written as:

$$\boldsymbol{e} = \boldsymbol{\Phi}\boldsymbol{w} \propto \frac{\widehat{q}_1}{\omega_1^2}\boldsymbol{\Phi}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^2}\boldsymbol{\Phi}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2}\boldsymbol{\Phi}\boldsymbol{p}_{k_1} \tag{A36}$$

where $\boldsymbol{\Phi}$ represents the $[m \times n]$ linear portfolio-to-exposure transformation matrix.

So the portfolio massaging problem (A35) reduces to finding ω_j^2 $(1 \leqslant j \leqslant k_1)$ such that:

$$\boldsymbol{e}^* = \frac{\widehat{q}_1}{\omega_1^2}\boldsymbol{\Phi}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^2}\boldsymbol{\Phi}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2}\boldsymbol{\Phi}\boldsymbol{p}_{k_1}. \tag{A37}$$

This m-linear system involves k_1 unknowns. If $m < k_1$ (i.e. views outnumber exposure criteria), the system is underdetermined, yielding infinitely many solutions that precisely achieve the ideal profile $\boldsymbol{e}^*$. Since investors are cognitively *unable to distinguish* view uncertainties but can *discern* exposures, any feasible set of ω_j^{*2} $(1 \leqslant j \leqslant k_1)$ suffices to recover the ideal portfolio:

$$\boldsymbol{w}^* = \frac{\widehat{q}_1}{\omega_1^{*2}}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^{*2}}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^{*2}}\boldsymbol{p}_{k_1}. \tag{A38}$$

Since exposure represents welfare, $\boldsymbol{w}^*$ constitutes a Pareto improvement over $\boldsymbol{w}$.

If $m = k_1$, the investor has exactly sufficient degrees of freedom to solve the system uniquely and match the target profile $\boldsymbol{e}^*$ precisely.

If $m > k_1$, the investor has insufficient degrees of freedom to solve the overdetermined system precisely. Nevertheless, portfolio massage allows the introduction of additional views when needed, so degrees of freedom are typically not binding. Even without adding views, Pareto improvements can still be attained by approximating the ideal profile, essentially solving the following problem:

$$\left((\omega_1^2, \omega_2^2, \ldots, \omega_{k_1}^2)^{\mathrm{T}}\right)^* = \underset{(\omega_1^2, \omega_2^2, \ldots, \omega_{k_1}^2)^{\mathrm{T}}}{\operatorname{argmin}} \left\| \boldsymbol{e}^* - \left(\frac{\widehat{q}_1}{\omega_1^2}\boldsymbol{\Phi}\boldsymbol{p}_1 + \frac{\widehat{q}_2}{\omega_2^2}\boldsymbol{\Phi}\boldsymbol{p}_2 + \cdots + \frac{\widehat{q}_{k_1}}{\omega_{k_1}^2}\boldsymbol{\Phi}\boldsymbol{p}_{k_1}\right)\right\|_2^2. \tag{A39}$$

In practice, cognitive ambiguity implies that investors typically start with only a vague ideal exposure profile $\boldsymbol{e}^*$. Consequently, problems (A35), (A37), and (A39) may be ill-defined, rendering precise and complex optimisation both unjustified and difficult to implement.

Instead, with analytics visualising portfolio exposures, manual *view-driven, feedback-responsive* portfolio discovery process offers a more intuitive and flexible process, enabling investors to incrementally elicit $\boldsymbol{e}^*$ and refine portfolios without explicitly specifying it upfront. Once investors identify a portfolio with satisfactory exposure profile, initially ambiguous views can be readily quantified and implemented.

Because view uncertainties are positive, portfolio massage fine-tunes view strength while preserving directional integrity, and therefore cannot harm alpha. By integrating analytical feedback, portfolio massage turns cognitive or subjective ambiguity into an advantage, enabling investors to pursue holistic *strategic* goals without compromising timely *tactical* view implementation, thereby producing Pareto improvements.

Beyond security- or portfolio-level views, the same argument applies to factor or security-specific views, since they can be efficiently mapped to their replicating portfolios (Proposition 4.1). ■